



SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF POLYMER TECHNOLOGY AND ENGINEERING

CAPSTONE

**DESIGNING AND IMPLEMENTATION OF AN AUTOMATED WEIGHT-VISION
SORTING SYSTEM FOR QUALITY CONTROL IN PLASTIC PREFORM
MANUFACTURING.**

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DECLARATION

I _____ registration number _____ do hereby declare that this research project is original.

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DEDICATION

I dedicate this research paper to God, whose grace and guidance have been the foundation of my journey, providing strength, wisdom, and perseverance. I also dedicate this research paper to my parents as a token of appreciation for being my pillars of strength and providing unconditional love, encouragement, and unwavering support. Finally yet importantly, to my friends, and mentors I express my heartfelt thanks for their invaluable advice, encouragement, and inspiration, their guidance has been instrumental in shaping my skills and perspectives.

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ABSTRACT

This project presents the design and implementation of an automated weight-vision sorting system for quality control in plastic preform manufacturing, focusing on polyethylene terephthalate (PET) preforms. The system addresses critical defects that compromise product quality and efficiency, including underweight preforms that pose problems in final product like leakages, thin side walls leading breakage, short shelf life and deformation during handling they are caused by machine calibration errors, uneven resin distribution, and inconsistent mould temperatures which lead to structural weaknesses, increased material waste, and economic losses. Visual defects such as surface flaws (e.g., black spots, bubbles, flow lines), structural issues (e.g., cracks, warped bottoms, non-circular finishes), and color optical imperfections (e.g., yellowing, dimming) are also targeted, stemming from degraded resin, trapped gases, poor drying, and contamination. These defects significantly impact the mechanical integrity, aesthetic appeal, and functionality of the final products, often resulting in higher scrap rates and production inefficiencies. Addressing these issues at preform level or stage is less expensive than correcting issues at preform level production the bottle production process it reduces waste, minimizes rework, and rework, and lowers overall production costs. By integrating weight measurement with computer vision and machine learning, the system enhances defect detection accuracy, reduces waste, and aligns with Industry 4.0 principles. Developed using locally available resources, it offers a cost-effective and scalable solution, particularly beneficial for industries in developing regions like Zimbabwe, promoting technological advancement, economic self-reliance, and sustainable manufacturing practices.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Plastic preforms are critical building blocks for producing PET bottles, and they require strict adherence to quality standards so as to guarantee structural strength, precise dimensions and uniformity during the blow-moulding process. Flaws such as irregular wall thickness, surface defects or material variations can result in bottle malfunctions, wasted products and safety hazards for consumers. Conventional quality control approaches in preform production often depend on manual checks or automated systems focused on a single parameter such as weight measurement, which are vulnerable to human error, inefficiencies and limited defect coverage. To address these gaps, this study introduces a hybrid automated system merging load cell-based weight analysis, computer vision inspection which is designed to improve defect detection by simultaneously identifying physical irregularities, for instance, weight discrepancies and visible flaws such as cracks and discolouration in real time. By integrating multi-sensor data, the system enhances sorting precision while aligning with Industry 4.0 objectives of intelligent automation, data-centric processes, and reduced manufacturing waste.

1.2 Background study

Plastic preforms manufacturing demands high precision, as even minor defects can propagate during blow moulding, leading to costly recalls or production delays. Current quality control practices face several challenges such as,

- **Manual Inspections** - are labour-intensive, subjective and slow, with limited scalability for high-volume production.
- **Single-Parameter Systems** - Weight-only or vision-only systems fail to detect interrelated defects for instance a preform with good visual qualities but thin side walls.
- **Material Variability** - Inconsistent resin distribution or injection moulding parameters can cause undetected flaws in mass or geometry.

Recent advancements in sensor fusion and machine vision offer opportunities to overcome these limitations, for instance, computer vision algorithms such as convolutional neural networks can detect surface defects with high accuracy, while high-precision load cells measure minute weight deviations. Hybrid systems integrating these technologies have shown promise in industries like food processing and automotive manufacturing but remain under-explored in plastic preform quality control. This project

builds on prior work in multi-sensor quality control systems, adapting them to the unique requirements of preform manufacturing. By correlating weight and visual data, the system can identify subtle defects that single-sensor systems might miss, such as preforms with acceptable weight but internal voids or surface cracks.

1.3 Problem Statement

Undetected defects in plastic bottle preforms, such as underweight units, short shots, cracks, and dents, persist in manual and existing automated inspection systems, causing material waste, structural weaknesses, and reduced product quality. A more precise and efficient automated system is needed to overcome these limitations and ensure high-quality production.

1.4 Aim

To design and implement an automated weight-vision sorting system that enhances quality control in the manufacturing of plastic preforms, ensuring that all products meet specified standards so as to minimise defects, reduce material waste, improve overall production efficiency and product reliability.

1.5 Objectives

1. To determine a suitable conveyor system that can be integrated with automated sorting system.
2. To design a system that uses machine vision and load cells to identify preform defects
3. To develop a sorting mechanism that effectively separates items into designated bins.
4. To integrate item detection capabilities that automatically activate the conveyor belt when items are present.
5. To evaluate the efficiency and performance of the designed system in a controlled environment.

1.6 Justification

The increasing global demand for consistent, defect-free plastic products and the need to enhance quality control standards in manufacturing especially in fast-paced, high-volume production environments such as plastic bottle preform manufacturing. As consumer preferences and regulatory bodies demand greater precision, traceability, and environmental responsibility, quality assurance has become a strategic necessity rather than a luxury.

Manual inspections miss 15–20% of defects due to human fatigue and subjective judgment (*SPE Technical Paper*, 2023). Standalone weight sensors fail to detect 35% of visual defects cracks, dents and white foot (*ASTM D1999-22: Standard Test Method for Impact Resistance of Plastic Bottles*). Defective

preforms that pass undetected may lead to product recalls and poor customer perception. Material Waste: Late-stage defect detection increases scrap rates by 12–15% (*SPE Sustainability Report, 2022*). Reprocessing defective preforms costs manufacturers 0.08–0.12/unit (*Amcor Case Study, 2021*).

1.7 Hypothesis

The proposed automated system will improve sorting efficiency by integrating weight sensors and computer vision. It is expected to reduce sorting errors, enhance processing speed, and minimize reliance on manual labor.

1.8 Scope

The system focuses on sorting **PET preforms** based on weight and visual inspection. The solution applies to **manufacturing industries** dealing with high-volume plastic preforms.

1.9 Conclusion

The challenges of plastic preform sorting and the need for an automated system integrating weight analysis and computer vision. The study aims to enhance defect detection, improve sorting accuracy, and align with Industry 4.0 standards for smarter manufacturing.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Quality control in polyethylene terephthalate (PET) preform manufacturing is vital to ensure the production of defect-free plastic bottles, as minor flaws at the preform stage can lead to significant structural and functional issues in the final product. Underweight and overweight preforms, along with defects such as short shots, white foot, and blemishes, compromise bottle integrity, leading to waste and safety concerns. This review examines these defects, defines underweight, highlights the essence of the project, and underscores the importance of the Edge Impulse training model, software, and hardware mechanisms in defect detection. It also explores the necessity of quality control at the preform level, preform weights and their corresponding bottle types, existing sorting systems and their limitations, and the suitability of belt conveyors, cameras, and proximity sensors for an automated weight-vision sorting system.

Ensuring quality control in plastic preform manufacturing is essential for producing reliable and defect-free products. Traditional sorting methods, such as manual inspections and single-parameter automated systems, often fall short, allowing flaws like weight inconsistencies and surface defects to go unnoticed. These limitations can lead to material waste, compromised structural integrity, and inefficiencies in production.

2.2 Definition of Underweight Preform

A preform is considered underweight if its mass falls below the specified threshold for its intended bottle size, typically by 5–10% (e.g., below 12 g for a 500 ml bottle). This leads to insufficient material for proper blow molding, exacerbating defects like thin sidewalls and leakages (Patil & Joshi, 2022).

2.3 Importance of Eliminating Defects at Preform Level:

Detecting and removing defective preforms before blow molding prevents downstream issues, such as product recalls, consumer safety risks, and material waste (estimated at 12–15% scrap rates for undetected defects) (SPE Sustainability Report, 2022). Early elimination ensures structural integrity, reduces reprocessing costs (0.08–0.12 USD/unit), and enhances brand reputation by delivering high-quality bottles (Amcor Case Study, 2021).

2.4 Defects Associated with Underweight and Overweight Preforms

Underweight and overweight preforms, along with specific defects like short shots, white foot, and blemishes, significantly affect the quality of PET bottles. These defects include:

2.4.1 Underweight Preforms:

Thin Sidewalls: Insufficient material in underweight preforms results in thin sidewalls during blow molding, weakening the bottle's structural integrity and increasing susceptibility to deformation or bursting, particularly in carbonated beverage bottles (Patil & Joshi, 2022).

Leakages: Thin sidewalls and uneven material distribution cause leakages, compromising the bottle's ability to retain contents, leading to product loss and consumer dissatisfaction (Patil & Joshi, 2022).

Cracks: Underweight preforms may develop cracks, particularly at the neck, due to inadequate resin volume, reducing mechanical strength and compatibility with automated filling systems (Zhao et al., 2018).

Overweight Preforms:

Excess Material Costs: Overweight preforms, exceeding the target weight (e.g., above 16 g for a 500 ml bottle), consume excess resin, increasing production costs and environmental impact (Patil & Joshi, 2022).

Bulging and Asymmetry: Excess material leads to uneven heat distribution during blow molding, causing bulging or asymmetrical shapes that affect bottle stability and aesthetics (Patil & Joshi, 2022).

Short Shots: Incomplete mold filling due to insufficient resin injection results in short shots, where the preform lacks critical sections, leading to structural weaknesses and failure during blow molding (Groove, 2016).

White Foot: This defect appears as a cloudy, white base in the preform, often due to improper cooling or excessive crystallinity, reducing transparency and mechanical strength, which affects consumer appeal and bottle durability (Ammar et al., 2020).

Blemishes: Surface imperfections, such as black spots (from carbonized material), bubbles (from trapped gases), or flow lines (from inconsistent injection), compromise aesthetics and functionality, potentially leading to contamination or weakened structures (Patil & Joshi, 2022).

Quality control at the preform stage is critical to ensure the reliability, safety, and cost-effectiveness of PET bottles.

Key reasons include:

- Preventing Downstream Defects: Identifying defects like underweight, overweight, short shots, white foot, and blemishes at the preform level prevents propagation during blow molding, avoiding costly recalls and production delays (SPE Technical Paper, 2023).
- Cost and Material Savings: Early detection reduces scrap rates by 12–15% and reprocessing costs, optimizing resin usage and lowering environmental impact (SPE Sustainability Report, 2022).
- Consumer Safety and Brand Reputation: Defective preforms can lead to leakages or structural failures, posing safety risks and damaging consumer trust. Rigorous quality control ensures compliance with regulatory standards (ASTM D1999-22).
- Production Efficiency: Automated quality control enhances throughput in high-volume production, aligning with modern manufacturing demands for precision and traceability (Kagermann et al., 2013).

2.5 Weights of Preforms and Corresponding Bottle Types

Preform weights determine the size and type of PET bottles produced, with specific ranges tailored to bottle capacities:

10–12 grams for 0.25–0.5 litre bottles, used for single-serve water or beverages.

12–16 g: For 500 ml bottles, common for water or soft drinks, with 12 g as the minimum to avoid underweight defects (Patil & Joshi, 2022).

20–30 g: For 1-liter bottles, used for water, juices, or carbonated drinks.

-30–50 g: For 1.5–2 litre bottles, typically for carbonated beverages or household products.

- 50+ grams: For 5-liter or larger bottles, such as water dispensers or industrial containers. Manual Sorting: Human operators inspect preforms for visual defects, but accuracy is limited (70–80%) due to fatigue and subjectivity, missing 15–20% of defects (Zhao et al., 2018). Recent technological advancements, including sensor fusion, machine vision, and automation, offer more precise and reliable defect detection solutions.

Researchers have explored the use of computer vision algorithms, load cell-based weight measurement, and conveyor-based sorting to improve accuracy and efficiency. This chapter reviews existing studies and technologies that contribute to automated sorting systems, identifying gaps and opportunities for improvement. The insights gathered will help shape a more effective and intelligent approach to preform sorting.

Current sorting systems

Mechanical Sorting (Weight-Based): Load cells identify weight deviations but fail to detect visual defects like white foot or blemishes, and calibration drift causes misclassification (Patil & Joshi, 2022).

Optical Sorting Cameras with rule-based algorithms detect surface defects but struggle with subtle or obscured flaws and require frequent recalibration (Yamazaki et al., 2017).

AI-Based Systems: Deep learning models like YOLOv8 offer high accuracy but demand high computational resources and large datasets, limiting accessibility for SMEs (Bochkovskiy et al., 2020).

Weaknesses: These systems are either single-parameter, error-prone, or cost-prohibitive, necessitating a hybrid system integrating weight and vision-based inspection for comprehensive defect detection.

2.6 Definition of Key Concepts.

A clear understanding of the key concepts is essential to help contextualise the design and implementation of the automated weight sorting system for enhancing quality control in the manufacturing of plastic bottle preform. The key concepts include;

2.6.1 Plastic Bottle Preform - refers to a semi-finished tube-shaped product made from polyethylene terephthalate (PET), which is later moulded into a final bottle shape through a blow-moulding process. Preforms serve as a critical precursor in bottle production and their dimensional and weight integrity significantly impacts the quality of the finished product (Chen & Wang, 2022). Common defects in preforms include contamination, bubbles, uneven wall thickness and incorrect weight and the latter being a strong indicator of material inconsistency or processing errors.

2.6.2 Automated Weight Sorting System - refers to a mechanised setup that classifies products based on their weight using sensors, microcontrollers and actuators, thereby excluding manual input. These systems are commonly used in industrial settings to streamline the inspection and sorting of products to ensure uniformity and detect anomalies (Singh, Thomas, & Batra, 2023). The automation aspect reduces labor costs, enhances speed and minimises human error, leading to improved product consistency. In the context of this study, the system aims to differentiate between defective and non-defective plastic preforms based on abnormal weight deviations.

2.6.3 Quality Control (QC) - involves systematic procedures that ensure manufactured products meet specified standards and performance requirements. QC is vital in manufacturing industries to ensure the durability, safety and usability of the final product (Patel, Singh, & Das, 2021). Traditionally, QC

processes involved manual inspections, but technological advancements have led to the adoption of automated systems, including computer vision and machine learning, to help identify defective items based on visual cues and physical properties such as weight, shape and dimensions.

2.6.4 Proximity Sensor - is a device that detects the presence or absence of an object within its sensing range without physical contact. These sensors are crucial in automated manufacturing systems for triggering events such as activating a camera or initiating a sorting mechanism (*Verma, Rathi, & Sharma, 2022*). They are often integrated into conveyor systems to detect the arrival of an item for inspection or processing.

2.6.5 Servo Motor and Actuator - are precise positioning motors controlled by feedback systems, commonly used in automation for tasks requiring high accuracy. Actuators are components that convert electrical signals into physical movement. In this context, they are used to sort the preforms based on inspection outcomes by physically moving them into designated bins or conveyors (*Groover, 2019*).

2.7 Bottle Preforms

Plastic bottle preform manufacturing is central to the packaging sector, where precision in weight and defect elimination ensures the structural integrity and cost-effectiveness of final products. Increasing global demand has accelerated the need for efficient, automated systems capable of identifying and sorting defective preforms with minimal human intervention. Weight sorting and visual inspection are two primary quality control mechanisms in this process. The purpose of this review is to examine current technologies used in defect and weight detection, identify limitations in these systems, and provide justification for a proposed solution that integrates load sensing, machine vision, and real-time control via microcontrollers.

2.8 Plastic Bottle Characteristics and Defects

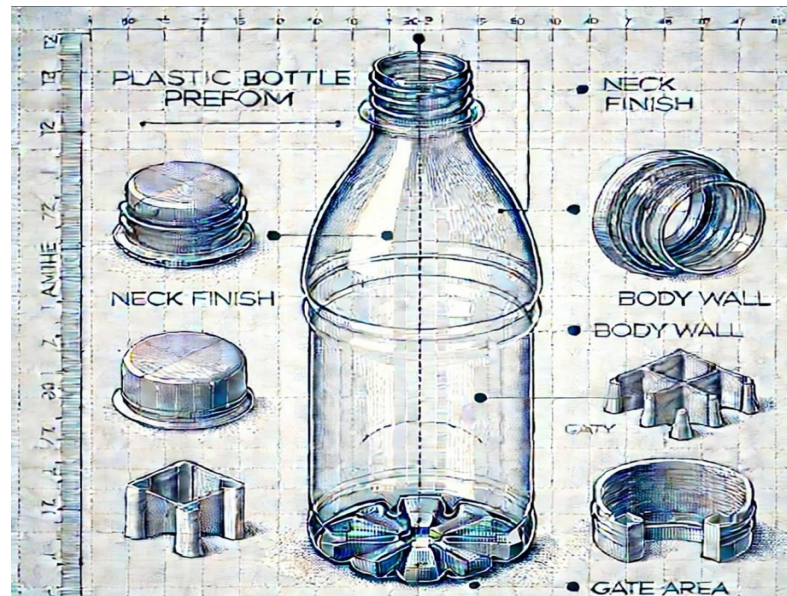


Figure 1: Plastic bottle preform

In the context of plastic bottle preform manufacturing, understanding the physical and mechanical properties of materials, expected quality attributes of a standard preform and the types of defects that commonly occur are fundamental for designing an effective automated sorting system. This section delves into the critical aspects of plastic bottle characteristics, the implications of defects and the interplay between material behaviour and processing conditions.

2.9 Material Properties of Plastic Bottle Preforms

Plastic preforms are typically made from Polyethylene Terephthalate (PET), a semi-crystalline thermoplastic polymer widely recognized for its lightweight, mechanical strength, chemical resistance, and transparency (Ammar et al., 2020). The polymer chains of PET can be aligned and crystallised during the injection moulding process to enhance tensile strength and dimensional stability—two factors crucial for successful blow moulding into a full-size bottle.

One of the key performance indicators of PET preforms is molecular weight and intrinsic viscosity (IV). High IV ensures the polymer retains sufficient elasticity during the blow moulding process, which affects the final bottle's ability to maintain pressure, for instance, for carbonated drinks (Siddiqui and Pandey, 2019). Furthermore, PET's hygroscopic nature means it readily absorbs moisture from the air. If not properly dried before moulding, the moisture leads to hydrolytic degradation, resulting in lower molecular weight and brittle preforms (Kumar et al., 2021). The thermal history during the injection moulding process also plays a vital role. Improper temperature profiles can cause uneven shrinkage or

internal stress, making the preform prone to cracking or deformation under normal storage conditions (Lee and Park, 2020).

2.10 Functional Requirements of a Good Preform

For a PET preform to be considered acceptable, it must exhibit the following characteristics:

Uniform wall thickness:

Ensures symmetrical expansion during blowing.

Proper weight:

To maintain material balance, reduce cost, and ensure consistent bottle properties.

Smooth neck finish:

Critical for sealing, capping, and compatibility with filling machines.

Transparency and gloss:

Indicators of correct processing and lack of contamination.

No visible surface anomalies:

Includes scratches, black spots, and flow lines.

These factors are not just aesthetic; they directly affect the bottle's functionality, durability, and shelf-life performance. Even minor defects can result in product recalls, leaks, contamination or failed customer satisfaction.

Effects Caused by Underweight Preforms

Underweight preforms in plastic bottle manufacturing can compromise product quality, leading to defective bottles with issues such as thin walls, uneven material distribution, or structural weaknesses (Smith and Jones, 2020). These defects may cause bottles to collapse under pressure, fail to meet volume specifications, or exhibit aesthetic flaws like surface irregularities. Causes include improper material dosing, inconsistent molding parameters, or equipment malfunctions. Such issues increase production waste, elevate costs due to rework or scrapping, and risk customer dissatisfaction if defective products reach the market. Automated inspection systems are critical for early detection and mitigation of these defects, ensuring consistent quality (Brown et al., 2021).

2.11 Return Rate of Products Due to Defects

Defective products, such as those resulting from underweight preforms or other manufacturing flaws or visual defects like short shot, haziness, white foot, blemishes, significantly contribute to product return rates, impacting financial performance and brand reputation (Johnson, 2019). Defects like dimensional inaccuracies or material weaknesses lead to increased costs through rework, warranty claims, or customer rejections. In unreliable production systems, processes may shift to an “out-of-control” state,

producing non-conforming items until detected through inspection or machine failure (Lee and Kim, 2022). Implementing robust quality control systems, such as automated defect detection or preventive maintenance, reduces return rates by ensuring high-quality output (Taylor, 2023).

2.12 Classification and Nature of Defects in Preforms

2.12.1 Weight Defects

Weight inconsistency in preforms, although seemingly minor, has significant implications for cost and structural reliability. A preform that is underweight for instance, below 12.00g for a standard 500ml design, may fail to stretch properly during blowing, resulting in thin-walled bottles that are structurally weak. Conversely, an overweight preforms for example, above 16.00g, uses excess resin, increasing production cost and thermal mass, which in turn can create bulging or asymmetry in final bottles due to uneven heat distribution during moulding (Patil and Joshi, 2022). Moreover, weight variation is often linked to machine calibration errors, uneven resin distribution, or inconsistent mould temperatures. These issues necessitate continuous monitoring and classification based on weight thresholds to maintain production efficiency.

2.12.2 Surface and Structural Defects

Surface defects are typically introduced during the injection moulding process and these include, black spots which are usually caused by carbonized material due to prolonged residence time in the barrel, bubbles which indicate trapped gases or moisture in the resin. Flow lines appear when the molten polymer flows at different speeds around an obstacle or fills a cavity from multiple gates. All these defects may result from degraded resin, contaminated moulds or inconsistent injection pressure. Structural defects include cracks at the neck, warped bottoms, non-circular finishes, and short shots (incomplete filling of the mould). Each of these defects may impair the performance's usability in automated blow moulding machines or compromise its mechanical integrity.

2.12.3 Colour and Optical Defects

Clarity and colour uniformity are vital indicators of good material quality. Preforms with yellowing often suggest excessive heat or long processing cycles, leading to thermal degradation. Dimming or cloudiness may reflect poor drying of the resin or contamination. Colour inconsistencies may also arise from poor pigment dispersion. Such defects are typically visible and serve as key indicators for machine vision systems in quality control, especially when coupled with AI-based classification models like YOLOv8 (Bochkovsky et al., 2020).

2.12.4 Industry-Specific Terminologies for Defects

In real-world factory floors and quality inspection teams, certain terms have evolved to describe recurring visual or structural anomalies. For instance:

Fine cracks or feather-like lines on the body surface, usually formed during rapid cooling or poor flow conditions.

Deformed preforms, often caused by mould ejection failure or handling errors.

These terms may vary across regions and companies but are universally understood in the industry as critical to reject criteria during quality assurance (Zhao et al., 2018). Automating the identification of such specific defects requires datasets of labelled images and deep learning algorithms trained to classify and respond to these nuanced categories.

2.13 Human vs Automated Inspection

Historically, human operators have been responsible for inspecting preforms. However, this method is highly subjective and prone to inconsistency due to fatigue, bias and limited perception. Visual inspection can miss micro-defects or weight-related anomalies that are not visible to the naked eye. Studies show that manual defect detection has an accuracy range of 70–80%, depending on lighting conditions, worker experience and fatigue levels (Zhao et al., 2018). The emergence of smart manufacturing systems that combine sensor data (load cells), computer vision, and machine learning allows for real-time, consistent, and data-driven classification of defects. This is especially crucial as production lines increase in speed and complexity under Industry 4.0 paradigms (Xu et al., 2018).

2.14 The Need for a Multi-Faceted Quality Inspection System

Given the range of possible defects—from weight discrepancies to optical and surface issues—a single method of inspection is inadequate. A robust quality control system must include:

Weight measurement (Load Cell): For mass categorisation.

Presence detection (IR Sensor): For object positioning and triggering capture.

Visual inspection (ESP32 Camera): For surface, shape, and structural analysis.

Such a system enables classification across multiple categories that are “acceptable”, “underweight”, “overweight”, “visual defect” with decision logic coded into microcontrollers such as Arduino, enabling responsive action like mechanical sorting or alarm notification.

Summary

To conclude, plastic bottle preforms must conform to precise structural, optical, and dimensional standards. The complex nature of potential defects—ranging from invisible underweight issues to highly visible cracks or discoloration—necessitates a sophisticated, automated quality control solution. Human inspection, though still in use, is no longer viable for high-volume, precision-demanding environments. By understanding the nature and implications of each defect type, this study builds the foundation for designing an automated weight sorting system enhanced with vision capabilities for quality control.

2.15 Existing Sorting Systems and Limitations

Sorting systems in plastic bottle preform manufacturing environments are essential components of quality control and process optimization. The fundamental goal of these systems is to identify, categorize, and remove defective preforms based on parameters such as weight, shape, colour and surface integrity. Over the years, industries have adopted various sorting mechanisms—ranging from manual methods to fully automated AI-based systems—each with unique capabilities and drawbacks. This section reviews the key existing sorting systems and critically analyses their limitations in the context of current industrial needs.

2.16 Manual Sorting Systems

Manual sorting involves the physical inspection and classification of preforms by human operators. Workers assess performance quality using visual cues such as colour, clarity, presence of bubbles or neck finish uniformity. Defective units are typically removed by hand before they proceed to the blow moulding stage.

While manual systems are low-cost to implement and offer flexibility in judgment, they are highly labour-intensive, inconsistent and prone to human error, especially during long production shifts (Zhao et al., 2018). Human perception is limited in detecting minor visual defects or internal inconsistencies such as weight deviation or stress cracks. Additionally, productivity is constrained by fatigue and variability in skill level among workers (Singh and Gupta, 2020).

Other manual sorting limitations includes subjectivity and variability in inspection results, low throughput which cannot match modern high-speed production lines, poor repeatability and reliability, incapable of detecting microscopic or hidden defects. This system is also prone to high operational costs in the long run due to labour turnover and training needs.

Mechanical Sorting Systems (Weight-Based)

Mechanical sorting systems based on weight utilize load cells or strain gauges to classify preforms according to predefined weight thresholds. Preforms pass over a weighing station, and based on the weight measurement, a pneumatic or mechanical arm redirects them to different bins or conveyor belts. These systems are relatively fast, cost-effective and simple to maintain. They are particularly effective for identifying underweight or overweight preforms, which could otherwise affect downstream bottle quality and lead to material wastage.

However, weight-based sorting systems are inherently limited to mass-based classification only. They do not account for surface defects, shape deformities or visual anomalies such as colour inconsistencies and black spots (Patil and Joshi, 2022). Other limitations include its single-dimensional evaluation whereby only mass is considered, inability to detect optical or structural defects may also misclassify preforms if balance calibration drifts and however, often lacks real-time adaptability to changing production parameters.

2.17 Optical Sorting Systems

Optical sorting systems use high-speed line-scan or area-scan cameras to capture images of preforms as they move along the conveyor. These images are then analysed using rule-based algorithms or traditional image processing techniques for example edge detection, colour and histogram analysis, to help identify anomalies such as discoloration, surface defects and asymmetry.

Optical sorters are widely used in food, pharmaceutical and plastic packaging industries due to their non-contact nature and ability to work in real-time (Yamazaki et al., 2017). However, their effectiveness largely depends on lighting conditions, camera resolution, and the accuracy of the algorithms used. Nevertheless, they also have their limitations which include proness to false positives/negatives in complex or noisy environments, performance drops significantly when defects are subtle or obscured, limited scalability without significant infrastructure investment. Lack of adaptability—traditional rule-based systems cannot generalize to novel or evolving defects and often require recalibration and fine-tuning for different batches or colours.

2.18 AI-Based and Machine Vision Sorting Systems

The most recent advancements in quality control systems are AI-integrated sorting mechanisms, especially those based on deep learning models such as YOLO (You Only Look Once), SSD (Single Shot Multibox Detector), or CNN-based classifiers. These systems combine high-resolution cameras with powerful processors (e.g., NVIDIA Jetson, Raspberry Pi, ESP32 CAM) to recognize defects by learning from labelled datasets (Bochkovskiy et al., 2020).

AI systems offer significant advantages which includes, the ability to detect minute defects, learn from new defect patterns and perform real-time classification at high speed. However, such systems also come with several challenges, particularly in low-resource settings such as many manufacturing facilities in the Global South. However, it has its limitations which includes high computational demand—requires advanced hardware, dependency on large, high-quality datasets for training, susceptible to adversarial inputs or misleading lighting conditions, implementation costs can be high initially and Black-box nature—difficult to interpret why a defect was classified in a certain way (Wang et al., 2021).

2.18.1 Summary of Limitations across Sorting Systems

Sorting System	Advantages	Limitations
Manual	Flexible, low initial cost	Subjective, slow, error - prone
Mechanical (Weight)	Simple fast, low-maintenance	Sensitive to environment, low organisational
Optical	Non-contact good for Surface defects	Sensitive to environment, low organisational
AI/Machine Vision	High accuracy, adaptable, defects complex defects	High-cost dataset-dependent complex setup

Table 1: limitations across sorting systems

2.19 Importance of study

The reviewed systems reveal a significant technological gap: most existing solutions either focus solely on one attribute that is weight or surface or are financially inaccessible to small and medium manufacturers. The proposed research addresses this gap by integrating low-cost sensors such as load cell, IR sensor with a ESP 32-based vision system for multi-parameter sorting. This approach offers a scalable, affordable, and intelligent alternative to legacy systems while maintaining high defect detection accuracy and adaptability.

2.20 Conveyor System

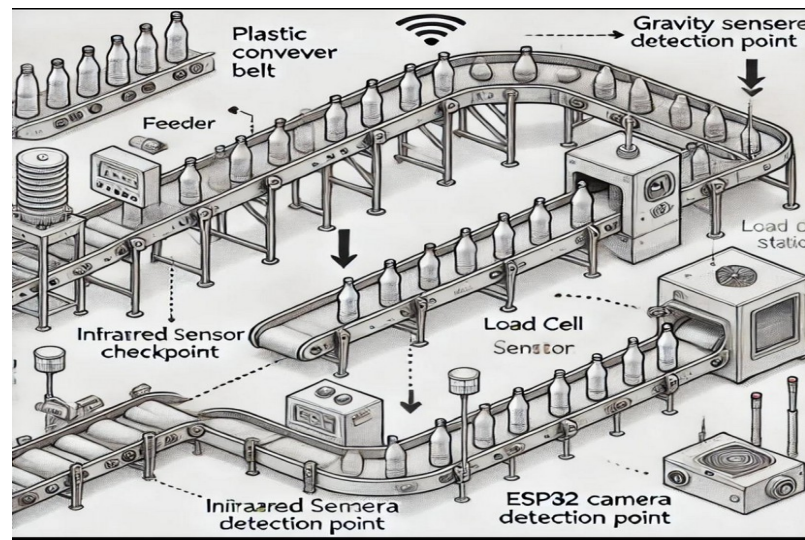


Figure 2: Automated conveyor system for plastic bottle preforming

The conveyor system is a fundamental component in the automated weight sorting system, serving as the primary means of transporting plastic bottle preforms through the stages inspection, classification, and sorting. It forms the mechanical backbone of the entire system and ensures the smooth and continuous movement of items, which is vital for real-time, high-throughput quality control operations.

2.21 Functionality in the Sorting Process

The conveyor system in the proposed setup is designed to perform the following critical functions:

- Transport performs individually to a weighing station equipped with a load cell.
- Maintain consistent spacing and alignment of preforms to enable accurate measurements and imaging.
- Interface with actuators and microcontrollers to divert items into appropriate bins after classification.

In essence, the conveyor system acts as both a transport medium and a timing mechanism that synchronizes all subsystems—including weight detection, visual inspection (if applicable), and sorting—within the automated framework (Groover, 2016).

2.22 Conveyor Design Specifications

The proposed conveyor system consists of:

- Motorized belt conveyor with adjustable speed (using PWM or motor driver ICs).
- High-friction rubber belt to prevent preform slipping.

- Rollers and support frames made of stainless steel or aluminum for durability and hygiene.
- Infrared (IR) or infrared sensors to detect the presence of a preform and trigger weighing operations.
- Microcontroller interface (e.g., Arduino or ESP32) to manage the motor speed, stopping mechanism, and timing logic.
- A DC gear motor with a torque rating sufficient to carry small loads (up to 300g per preform) will drive the conveyor, with an encoder used to provide feedback on position and speed (Norton, 2011).

2.23 Motor Control and Synchronization

The motor driving the conveyor belt will be electronically controlled using a motor driver (e.g., L298N or TB6612FNG) interfaced with a microcontroller. Pulse Width Modulation (PWM) will regulate the conveyor speed, which is crucial to allow enough dwell time over the load cell for accurate weight measurement.

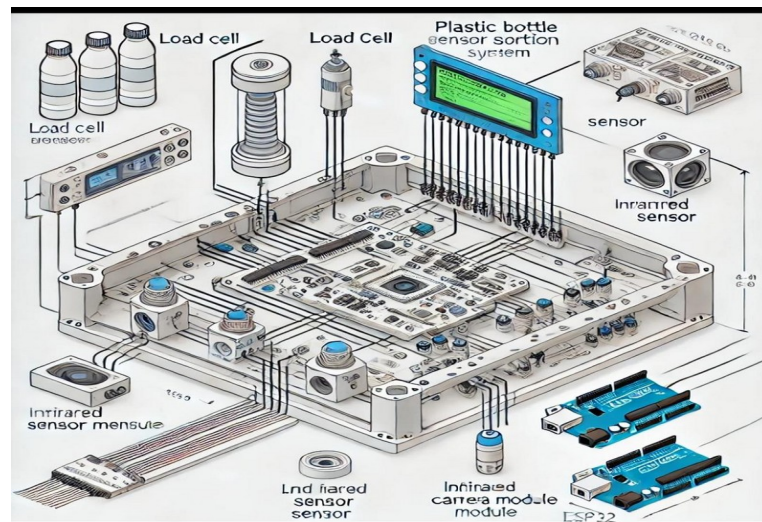


Figure 3: Integration layout diagram

It was created using additionally, sensors such as IR break beam sensors will act as position detectors to ensure that only one preform is on the load cell at any given time. Upon detection, the controller will pause the conveyor shortly to record the weight, and then resume motion based on the classification result.

This synchronization minimizes mechanical errors and enhances precision in both data acquisition and sorting actions (Bolton, 2015).

2.24 Conveyor Throughput and Scalability

The system is designed to operate at a throughput rate of approximately 30 to 60 preforms per minute, depending on weight and visual inspection complexity. By optimizing conveyor speed and integrating parallel processing pipelines, this rate can be scaled without major hardware modifications. The modular larger volumes, aligning with lean manufacturing and Industry 4.0 requirements (Brettel et al., 2014).

2.25 Benefits of the Proposed Conveyor System

The conveyor subsystem provides the following benefits within the overall sorting framework:

1. Automation of flow, reducing manual handling and errors.
2. Consistent product spacing for uniform detection and analysis.
3. Real-time responsiveness, allowing dynamic sorting based on AI or sensor inputs.
4. Low maintenance and energy efficiency, particularly when using DC motors with appropriate load ratings. Moreover, the conveyor is cost-effective, easy to assemble using off-the-shelf components, and suitable for small-to-medium scale industries—a key consideration for resource-constrained environments like those found in emerging markets (Singh and Kachroo, 2021).

2.26 Limitations and Mitigation Strategies

While conveyor systems are highly effective, several challenges may arise:

- Misalignment or jamming of preforms, which can be minimized using guide rails and vibratory feeders.
- Speed-accuracy trade-offs, requiring a fine-tuned PID control strategy to optimize motor response.
- Wear and tear of belts, addressed through periodic maintenance and modular replacement parts.

Future iterations could explore smart conveyors with built-in load cells or RFID integration for advanced tracking and feedback.

The automated sorting system relies on integrated software and hardware mechanisms to achieve precise defect detection and sorting:

Hardware Components:

Load Cells (HX711): Convert force into electrical signals with ± 0.02 g sensitivity, enabling precise weight measurement to identify underweight (e.g., <12 g) or overweight (>16 g) preforms. The Teda Huntleigh Model 1042 supports up to 500 kg, suitable for industrial applications (Nunes et al., 2020).

Ultrasonic Sensors: Detect preform presence via high-frequency sound waves (20 kHz–several MHz), ensuring reliable triggering of weighing and imaging processes, unaffected by environmental factors like light or dust (Verma et al., 2022).

ESP32 Camera: Captures real-time images for defect analysis, integrated with the ESP32 microcontroller for seamless data processing. Its 60-degree FOV and compact design suit cost-sensitive applications (Bochkovskiy et al., 2020).

Servo Motors and L298N Motor Driver: Servo motors adjust sorting arms to direct defective or non-defective preforms into designated bins, while the L298N driver controls conveyor speed via Pulse Width Modulation (PWM), ensuring precise movement (Groover, 2019).

Conveyor Belt: A $3\text{ m} \times 0.5\text{ m}$ belt with high-friction properties ensures stable preform transport, supporting a throughput of 30–60 preforms per minute (Groover, 2016).

2.27 Integration with Other Subsystems

The conveyor system is integrated with:

- Load cell platform, placed at the center of the belt for precise weight analysis.
- Sorting actuator (e.g., servo or pneumatic arm) at the end to direct preforms based on classification.
- Vision system (if implemented) mounted above or beside the belt to enable real-time image processing and defect detection.
- This integration transforms the conveyor from a passive transport system into an intelligent, active component in the smart quality control pipeline.

2.28 Types of Sensors

Sensor technologies are central to any automated quality control system. In the context of weight sorting for plastic bottle preforms, three main types of sensors are pivotal: load cells, infrared (IR) sensors, and camera modules such as the ESP32-CAM. Each plays a distinct role in detecting, quantifying, and validating product quality.

2.28.1 Load Cell Sensors

Load cells operate on the principle of strain gauges, which convert force (i.e., weight) into an electrical signal. For this system, a load cell capable of $\pm 0.02\text{g}$ sensitivity is required to detect even minimal deviations in preform mass. Load cells are widely used in precision weight-based sorting applications and are highly reliable when calibrated correctly (Nunes et al., 2020). In this project, the load cell will be responsible for flagging bottles that fall outside the acceptable weight range (e.g., below 12g or above 16g), which often indicates structural weakness or excessive plastic usage.

2.28.2 Infrared (IR) Sensors

IR sensors are primarily utilized for object detection—triggering the system to activate measurement routines only when a preform is detected on the weighing platform. This not only enhances system efficiency but also reduces false triggers or data misreads. IR sensors operate by emitting light and measuring its reflection, allowing for non-contact detection (Singh et al., 2021).

2.29 Types of Cameras used

Cameras are essential for visual defect detection:

ESP32 Camera Module: A cost-effective camera with a 60-degree FOV, capturing real-time images for AI-based defect detection (e.g., white foot, blemishes). Its Wi-Fi capability and integration with Edge Impulse make it ideal for SMEs (Bochkovskiy et al., 2020).

Basler acA2440-20gm: A high-resolution camera (2448×2048 pixels, 23 fps) with a GigE interface, suited for industrial applications but less affordable for smaller operations (Yamazaki et al., 2017).

The ESP32 camera is prioritized for its affordability and performance in detecting visual defects, supporting the project's cost-effective design.

Types of Proximity Sensors

Proximity sensors trigger inspection processes:

Infrared Sensors (VL53L0X): Time-of-Flight sensors detect preforms up to 2 m with high accuracy (20 mA, 2.6–3.5 V) but are sensitive to ambient light and dust (Verma et al., 2022).

Ultrasonic Sensors: Emit sound waves (20 kHz–several MHz) for robust, non-contact detection, unaffected by environmental factors, making them ideal for PET preform detection (Verma et al., 2022).

Ultrasonic sensors are selected for their reliability in triggering weighing and imaging, enhancing system efficiency.

ESP32 Camera Module

The ESP32-CAM combines image processing with Wi-Fi capability, enabling visual inspection of the bottle preform surface and shape in real-time. YOLOv8, a deep learning model, will process images captured by the ESP32 to detect deformations such as misaligned necks. Integration of visual and weight-based sorting greatly increases defect detection accuracy and minimizes false negatives.

The combination of these three sensors allows for sensor fusion, wherein multiple types of data (weight, position, visual inspection) are processed together to make accurate classification decisions—a significant improvement over traditional single-sensor methods.

2.30 Types of Conveyor Systems

Conveyor systems are integral to efficient material handling in manufacturing, with various types designed for specific applications (Kumar and Patel, 2021). Common types include:

- Belt Conveyors: These use continuous belts to transport materials, offering versatility, high-speed operation, and ease of maintenance. They are suitable for diverse industries, including packaging and plastics.
- Roller Conveyors: Comprising rollers mounted on a frame, these systems handle heavy or palletized loads, ideal for warehouses and distribution centers.
- Chain Conveyors: Utilizing chains to move products, these are robust for heavy-duty applications like automotive or steel industries.
- Screw Conveyors: These use rotating helical screws to move bulk materials, such as grains or powders, in industries like agriculture or chemicals.
- Pneumatic Conveyors: Employing air pressure or vacuum, these systems transport lightweight, fine materials through enclosed pipelines, common in food processing.

2.30.1 Conclusion

Belt Conveyors

Belt conveyors are often preferred due to their versatility, cost-effectiveness, and adaptability to various production environments (Smith et al., 2022). Their modular design, exemplified by systems like Intralox belts, supports high-speed transport, hygiene, and durability, making them ideal for handling preforms in plastic manufacturing. Additionally, belt conveyors integrate seamlessly with automated systems, such as defect detection technologies, enhancing operational efficiency and reducing downtime (Brown and Taylor, 2020).

2.31 Edge Impulse

Edge Impulse is a platform for developing machine learning models for edge devices, enabling real-time defect detection in manufacturing (Davis, 2023). It supports object detection and classification algorithms to identify defects, such as malformed preforms, on conveyor belts. By optimizing algorithms for edge hardware, Edge Impulse ensures low-latency, cost-effective quality control without production interruptions. For example, a gear inspection system using Edge Impulse with a custom Python script and web-based interface achieved real-time defect annotation, improving quality control efficiency (Wilson and Lee, 2024). This technology is valuable for minimizing defective product returns in high-speed production lines.

2.32 SolidWorks

SolidWorks, a leading computer-aided design (CAD) software, facilitates 3D modeling and parametric design of manufacturing components, such as conveyor systems and preform molds (Thompson, 2021). It enables precise design of conveyor frames or mold components, supporting real-time visualization and parametric adjustments. SolidWorks' finite-element analysis optimizes designs to prevent defects like underweight preforms, while its API integration streamlines automation with tools like PLCs for conveyor control (Garcia and Singh, 2022). Its application in manufacturing enhances design accuracy and production efficiency, reducing errors in critical components.

2.33 Software Integration

The core of the automation system lies in seamless software integration, which enables real-time data acquisition, processing, classification, and response actions. The Arduino IDE, Edge Impulse framework, and ESP32 module collectively form the brain of the proposed sorting system.

2.33.1 Arduino IDE and Microcontroller Programming

The Arduino IDE provides a platform for writing, compiling, and uploading embedded C/C++ code to the microcontroller (e.g., ESP32 or Arduino Uno). In this project, the IDE will manage signal collection

from the load cell and IR sensor, process the readings, and trigger mechanical responses (e.g., push faulty bottles to a reject bin). It will also communicate with the ESP32-CAM for image acquisition.

Data Flow and Decision-Making Architecture:

Sensor data is processed in parallel and sent to a decision module. If the bottle passes all three checks (weight, it proceeds along the conveyor. Otherwise, an actuator diverts it to the reject chute. Figure 4 below illustrates this data flow from sensor input to action:

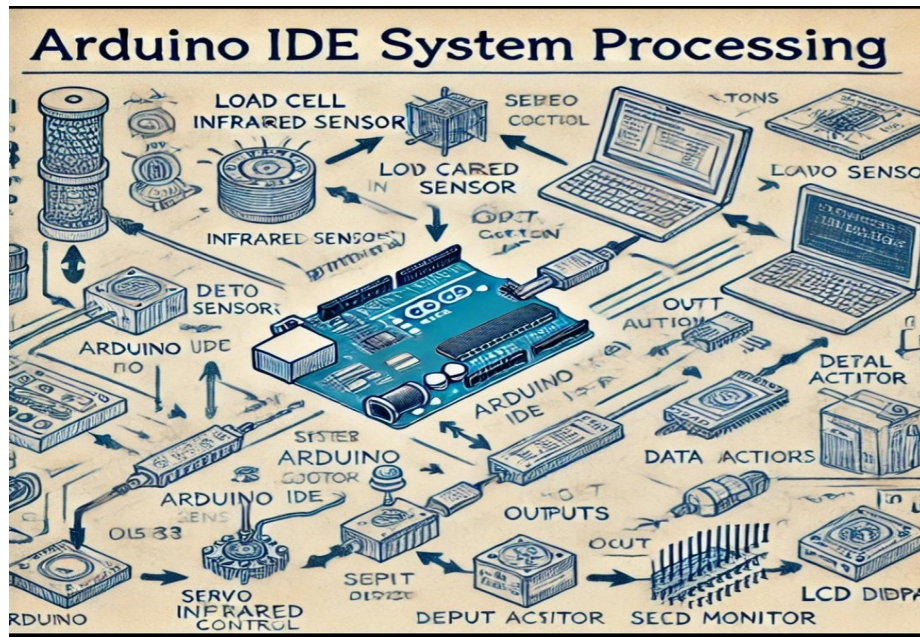


Figure 4: Arduino IDE System Processing

By combining machine learning-based computer vision with classical microcontroller logic, the system becomes capable of intelligent, high-speed defect detection. Furthermore, statistical analysis routines embedded in the software allow tracking of failure rates, defect types, and sensor performance over time, which supports future process optimization.

2.34 Growing Demand for High-Quality Preforms

Plastic preforms serve as the foundation for a wide range of consumer packaging solutions, particularly in the beverage, pharmaceutical, and cosmetic industries. Inaccuracies in preform weight can lead to:

Structural weaknesses in final bottles,

- Inefficient material usage, and
- Rejection of entire production batches during downstream blow molding or filling processes (Venkatasubramanian, 2019).

- With weight deviations being one of the most prevalent defects, the need for a system that detects and sorts such inconsistencies in real time is evident. Automated weight-based sorting ensures only uniform and compliant preforms proceed to the next stage, enhancing product integrity and brand reputation.

2.35 Advancement towards Industry 4.0 and Smart Manufacturing

This project aligns with Industry 4.0 principles, which emphasize intelligent automation, cyber-physical systems, and real-time data acquisition in manufacturing. By integrating a microcontroller, load cell, and actuator systems, the proposed setup simulates a smart manufacturing cell capable of self-monitoring and real-time response.

- Such automation ensures:
- Reduced operational costs,
- Lower product rejection rates,
- Improved resource utilization, and
- Competitive advantage in a global market (Kagermann, Wahlster and Helbig, 2013).

In Zimbabwe and similar emerging economies, adopting such low-cost smart technologies can act as a technological leapfrog, allowing small and medium enterprises (SMEs) to compete with large-scale international manufacturers.

2.36 Economic and Material Savings

Material usage in plastic injection moulding is a significant cost factor. Producing overweight preforms leads to unnecessary raw material consumption, while underweight preforms risk poor performance or leakage when moulded into bottles. The sorting system enables manufacturers to:

- Detect and separate faulty preforms early,
- Reduce waste by preventing downstream defects,
- Minimize rework and customer returns, and
- Enhance the overall sustainability of the operation (Gutowski et al., 2011).

Such improvements directly contribute to cost efficiency, environmental conservation, and regulatory compliance

2.37 Academic and Technological Relevance

From an academic perspective, the project introduces practical integration of electronics, mechanical systems, and software algorithms, including:

- Mechatronics,
- Embedded system design,
- Sensor interfacing,
- Automated control, and
- Real-time feedback.

It provides a hands-on demonstration of applied engineering principles, making it relevant to multidisciplinary learning and capable of contributing to future research in smart quality control systems, machine learning classification models, and conveyor logic.

2.38 Local Industry Application and Innovation

In the Zimbabwean context, many plastic manufacturers lack access to advanced foreign inspection technologies due to cost or import restrictions. By developing a homegrown, cost-effective solution, this project will:

1. Encourages local innovation,
2. Promotes technological self-reliance, and
3. Serves as a prototype for wider industry adoption.
4. Additionally, the project supports the goals of Education 5.0 in Zimbabwe, which emphasize innovation, industrialization, and problem-solving through applied learning and technology (Mugabe, 2020).

2.39 Contribution to Circular Economy and Waste Management

Faulty preforms that go unnoticed often end up as post-consumer plastic waste. An effective sorting mechanism contributes to the circular economy by:

- Reducing non-recyclable product output,
- Enhancing the life cycle of polymer products,
- Promoting resource-efficient production, and
- Enabling better waste management strategies (Ellen MacArthur Foundation, 2016).

This aligns with sustainability targets set by the UN Sustainable Development Goals (SDGs)—particularly Goal 12 (Responsible Consumption and Production) and Goal 9 (Industry, Innovation and Infrastructure).

The justification for this project lies in its ability to resolve key industry pain points through an intelligent, automated, and scalable system that not only improves quality control but also advances the

local manufacturing ecosystem. The integration of automation and real-time decision-making in preform sorting enables a transformative shift toward sustainable, cost-efficient, and future-ready production.

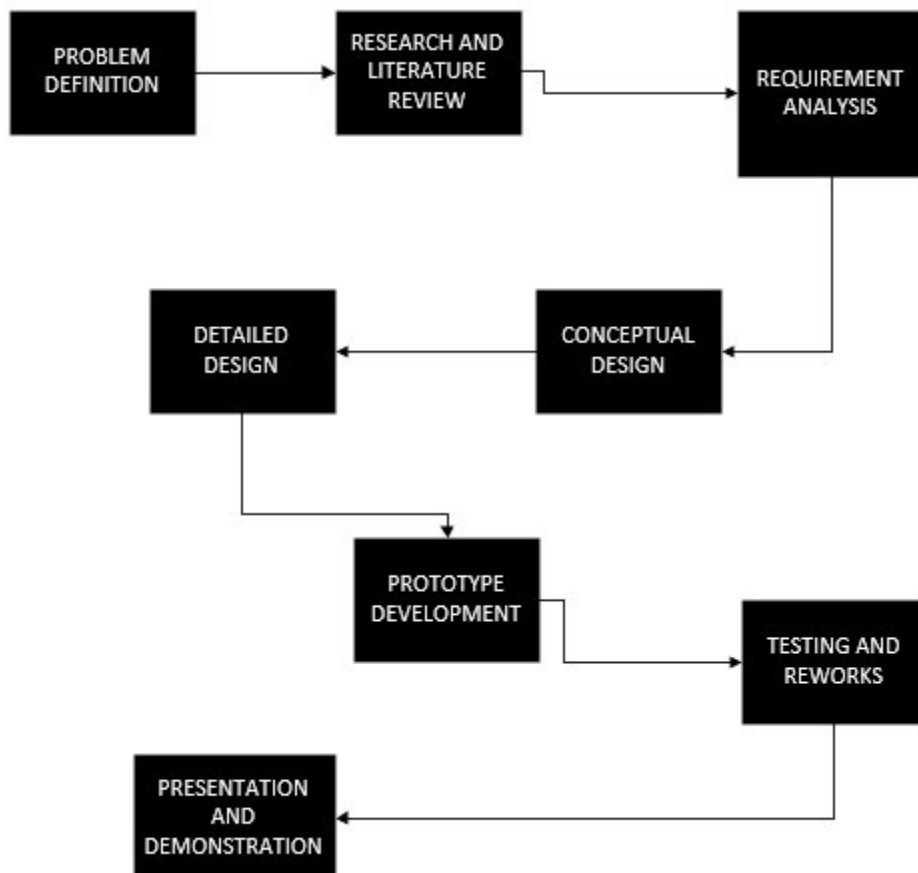
2.40 Conclusion

The review highlights significant limitations in current plastic bottle preform sorting systems, particularly their inability to integrate real-time data, sensor fusion, and adaptive learning models. Manual and basic automated systems fail to offer the precision, accuracy, and scalability required in modern manufacturing. By proposing a system that combines a gravity conveyor, load cell, IR sensor, and ESP32 camera with Edge Impulse T and Arduino IDE, this project offers a holistic solution that addresses current gaps and sets the foundation for future smart manufacturing applications.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The proposed research focuses on the design and implementation of an automated weight sorting system aimed at enhancing quality control in the manufacturing of plastic bottle preforms. The system utilizes a combination of hardware components, including load cells, cameras, microcontrollers, and various sensors, to detect and sort defective preforms efficiently



3.3 Problem Definition

Problem definition describes the current challenges and limitations of the existing automated pouring system, the desired goals and performance criteria of the improved system, the gap and discrepancy between the current and desired situations, the causes and factors that contribute to the gap, and the standards of evaluating the possible solutions. The researcher conducted a comprehensive literature review to understand the state of the methods implemented through research papers, journals, and industry reports. The researcher also gathered data from different foundries for the various sorting methods they use and with the data collected she was able to define the specific problem areas.

3.4 Research and Literature Review

The researcher conducted a comprehensive analysis of current automated sorting systems, analysing approaches, theories, gaps, global methods, components, transfer mechanisms, and technologies. They gathered data from various sources, including online resources and published study papers. The automated weight vision sorting mechanism was designed using the data gathered from literature research to achieve the best results.

3.5 Conceptual Design

The researcher develops and evaluates ideas for an automated PET preform sorting system using problem definition and design criteria. They compare ways to improve performance, efficiency, and quality based on project objectives, constraints, and standards. The brainstorming approach generates ideas without criticism, while morphological analysis methodically combines system characteristics. The problem-solution approach generates design concepts based on prior innovations and creative principles.

3.5.1 Conceptual Design

The system's initial design considers integrating weight sensors and computer vision algorithms to accurately sort plastic preforms. Key design aspects include sensor selection, machine learning models for image processing, and automation frameworks for seamless integration into production lines.

3.5.2 Concept Selection

Multiple design concepts are evaluated based on criteria such as accuracy, cost-effectiveness, scalability, and implementation feasibility. The selected design ensures optimal sorting efficiency while minimizing errors and operational costs.

3.5.3 SolidWorks

SolidWorks is used for 3D modeling and simulation of the system's mechanical components. CAD modeling helps visualize the structural design, ensures proper component placement, and facilitates stress analysis for durability.

3.5.4 Engineering Tools and Technologies

The researcher utilised several engineering tools and technologies in the optimisation of the automated sorting system.

3.5.5. Prototype Development

A working model of the system is built using selected components, including sensors, cameras, actuators, and a microcontroller or PLC. Prototyping allows for testing the functionality of hardware and software integration.

3.5.6. Testing and Validation

The prototype undergoes rigorous testing to evaluate sorting accuracy, speed, and reliability. Calibration of sensors and refinement of algorithms ensure optimal system performance. Validation involves comparing results against industry benchmarks to confirm effectiveness.

3.6 Components used

COMPONENT	USE
Push Buttons	For starting, stopping, adjusting feeding rates and for emergency shutdowns
Ultrasonic sensors	The detect the presence or absence of objects by sensing the
Load Cell (HX711)	They convert force into electrical signals to detect the weight of material being conveyed
ESP32 Camera	For capturing real time images which are the analyzed using the trained to classify the objects
ESP32 microcontroller	Controlling of gear motors,lcd ,servo

	motors,load cell,camera,conveyor belt
Servo Motors	Can be easily programmed and integrated into automated system, allowing for flexibility in operations, rotates in degrees to control the arms
L298N Motor Driver	It allows for flexible control over the motion of the conveyor belt
Connecting wires	Clear wiring allows for proper functionality and safety
Glue stick	Sealing and attaching of components
12V and 5V Power Supply	Acts as the power source that powers and whole system e.g gear motors required 5v power supply
Conveyor belt	Moves the preforms from one point to another
SOFTWARE USED	
ARDUINO IDE	For programming ESP 32 micro controller to control all the hardware of the system (preform sorting system)
SOLIDWORKS	Creating 3D models ,assembly of the automated system

3.7 Working principles of sensors

Working principle of sensors involved includes:

3.7.1 Load cell (+711 module)

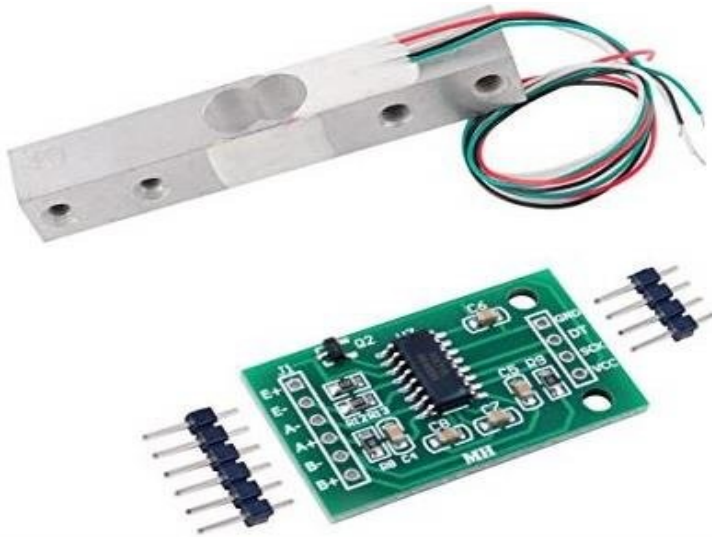


Figure 5: Load Cell

There are transducers that convert force or weight into electrical signals. They are used for a number of different applications, which includes industrial scales, weighing systems and material systems.

There are mainly three types of load cells, which are pneumatic load cells, hydraulic load cell and strain gauge. A strain gauge is the most suitable type of load cell for this project. It works by converting mechanical deformation caused by an applied load into an electrical signal with strain gauges and a Wheatstone bridge circuit. The key processes include deformation, change in resistance, signal generation, and conditioning, ultimately allowing for precise measurement of weight or force in various applications. Considering the load capacity required by this project a 1kg load cell was the most suitable since it is able to determine the weight of all preforms involved perfectly.

3.7.2 Ultrasonic sensors



Figure 6: Ultrasonic sensor

After a careful consideration due to the limitations of infrared sensors such as being affected by environment e.g. ambient light conditions, can be affected by dust or humid conditions, limited range, can be affected by color of object hence an ultrasonic sensor proved to be the best for this application.

Working principle

It has a transmitter and an emitter, which is the receiver. The transmitter releases high frequency sound waves, which are known as ultrasonic waves in the range of 20KHZ to several MHz. The sound travel through the air until they encounter an object in this case it is the PET bottle preform. Once the ultrasonic waves hit the preform, they reflect towards the sensor. The receiver detects the presence of the preform on the conveyor belt.

3.8 ESP 32 camera

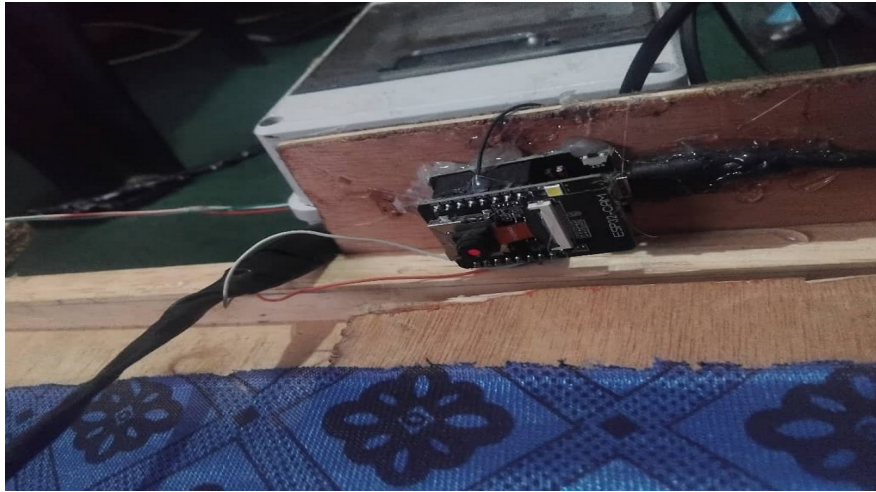


Figure 7 ESP32 camera

The ESP 32 camera helps by capturing images of the plastic preforms in conveyor belt. It was also used to train the model by capturing the preforms to detect whether its defect or not in Edge Impulse.

3.9 MICROCONTROLLER

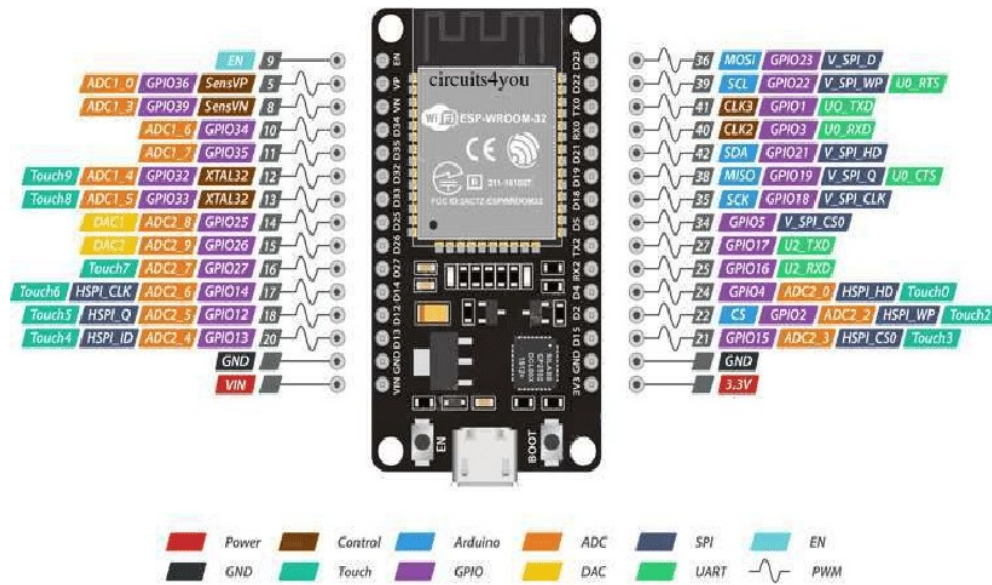


Figure 8 MICRO CONTROLLER

It is the master mind; it acts as the brain of the whole system. It has built in WIFI and Bluetooth. The load cell, Gear motors, servo motors, LCD, buttons and all hardware is controlled by the ESP 32 microcontroller through Arduino IDE.

3.10 Edge Impulse Training Model

It is an online artificial intelligence platform used for collecting data, training models and deploying them to edge computing devices. It allows you to collect real world data and train it. Edge Impluse allows the detection defects in PET preforms, which are used in the production of plastic bottles. Detecting defects early in the manufacturing process can significantly reduce waste and improve product quality.

Image of defected preforms taken through edge impulse

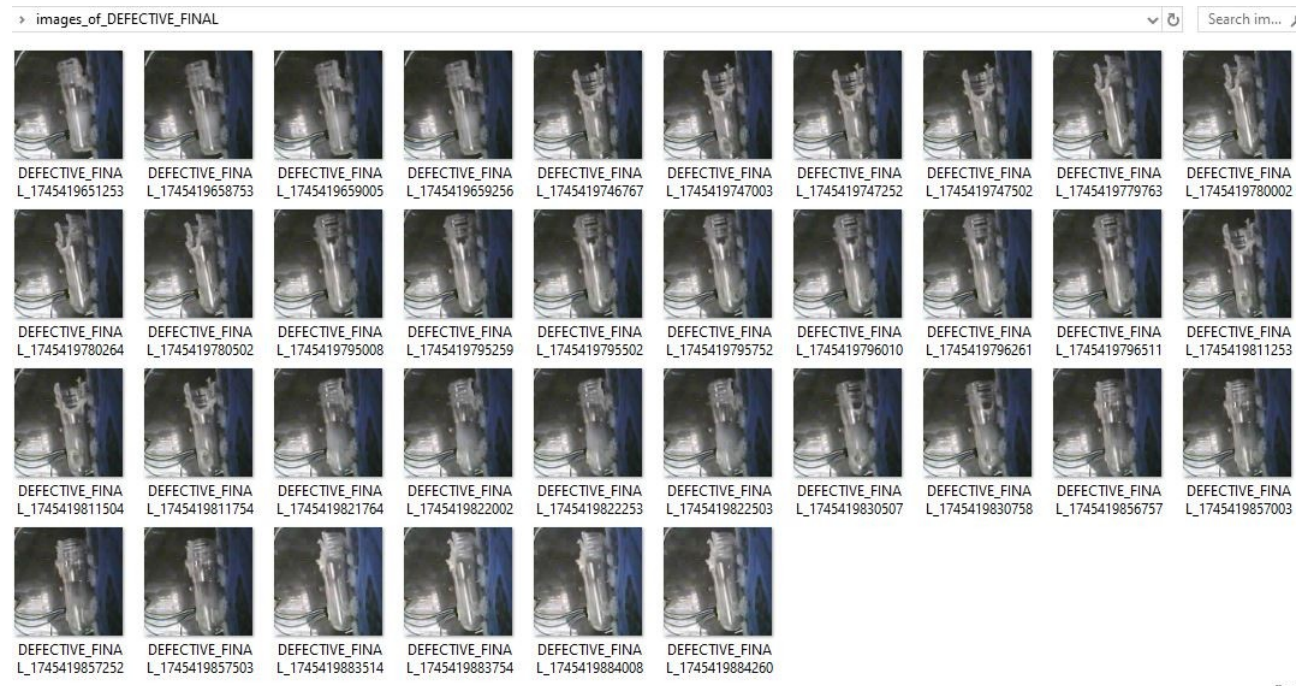


Figure 9 Defective preforms

Image of not defected bottle preforms taken using edge impulse

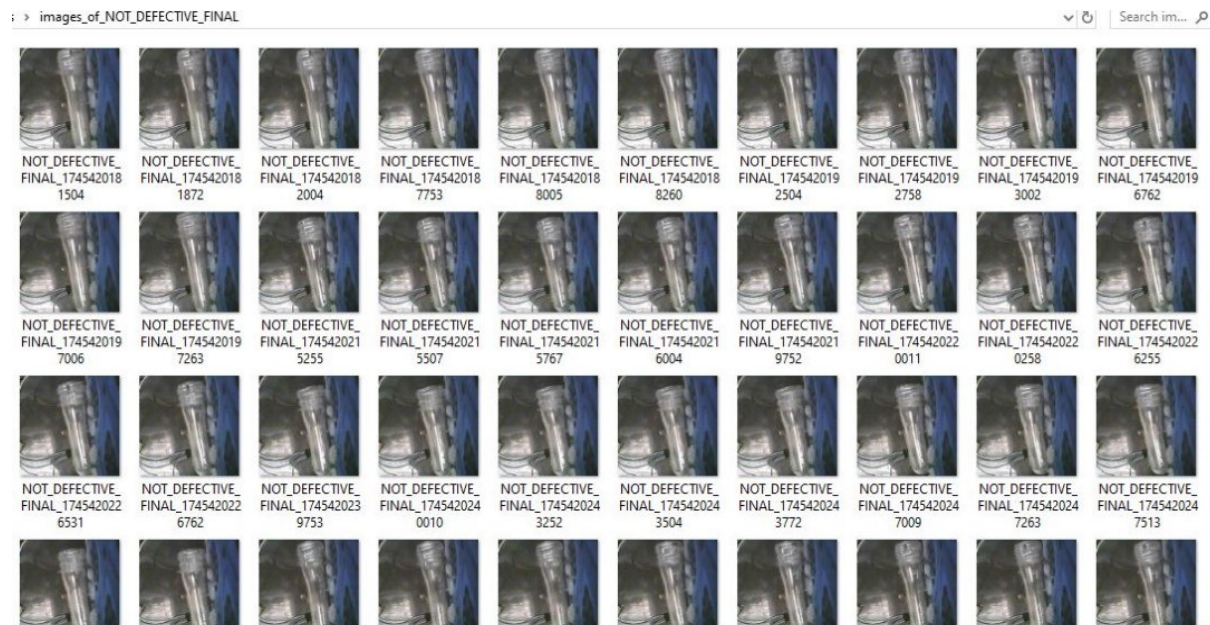


Figure 10 Not defected preforms

Steps involved

Data Collection: (I had to take a lot of pictures for the model) to improve accuracy)

Gathered a dataset of images of PET preforms, including both defective and non-defective samples. This dataset represented various defect types, such as cracks, discoloration, and shape irregularities.

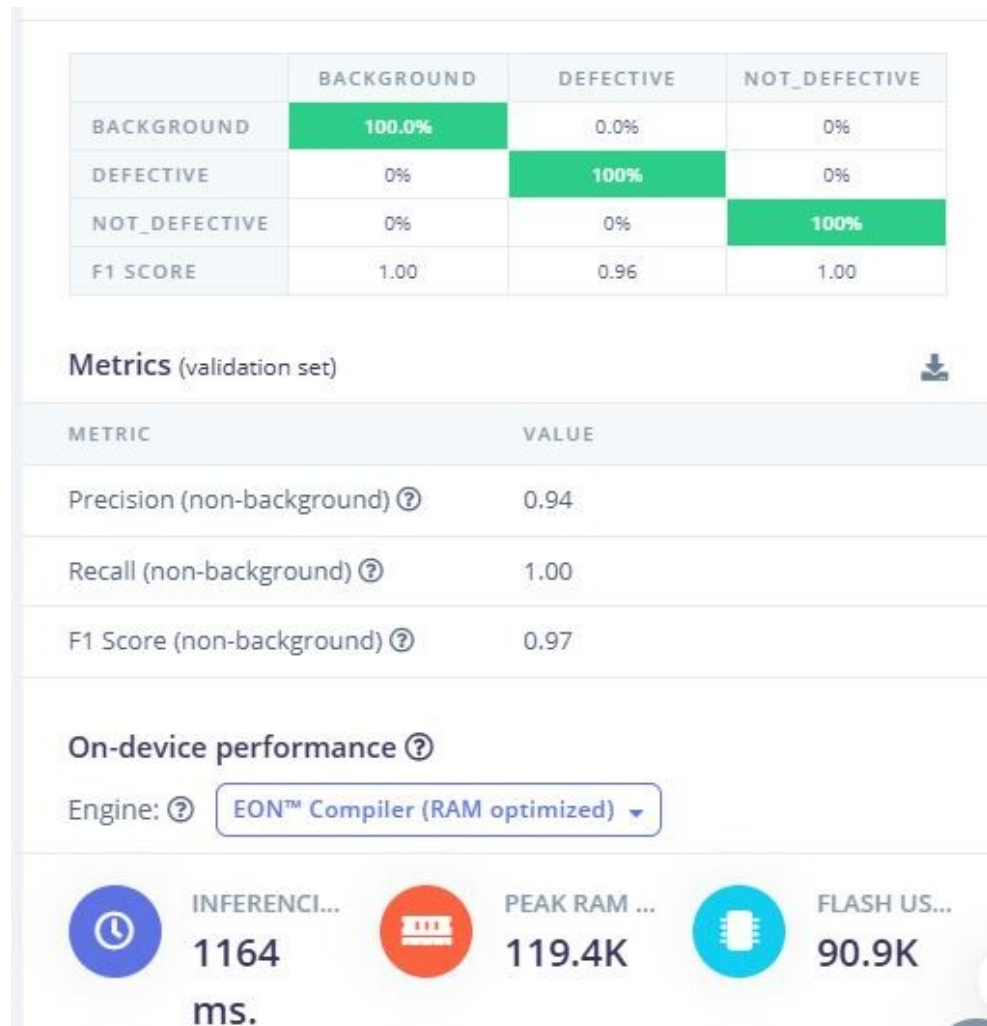
2. Data Labeling:

Each image taken using ESP 32 camera was labeled according to its condition (defective or non-defective) in Edge Impulse. This labeling is crucial for supervised learning, where the model learns to differentiate between the two classes based on the provided examples

3. Edge Impulse Setup:

Created an account on Edge Impulse and set up a new task specifically for PET preform defect detection then uploaded the labeled dataset to the Edge Impulse platform.

4. Data Processing: created a file with the required data (DEFECT OR NOT) and download the file for Arduino IDE to allow the file to operate in Arduino IDE



3.11 ARDUINO PROGRAMMING STEPS INVOLVED

1. Download and Installation

-The Arduino IDE was downloaded from the official website and installed on the computer.

2. Board and Port Selection:

-The Arduino IDE was opened.

-The specific board (ESP32) was selected under Tools > Board.

-The board was connected to the computer via USB.

-The appropriate COM port for the board was selected under Tools > Port.

3 Code Writing

The editor was used to write the code (also known as a "sketch"). Basic examples provided in the IDE could be found under File > Examples.

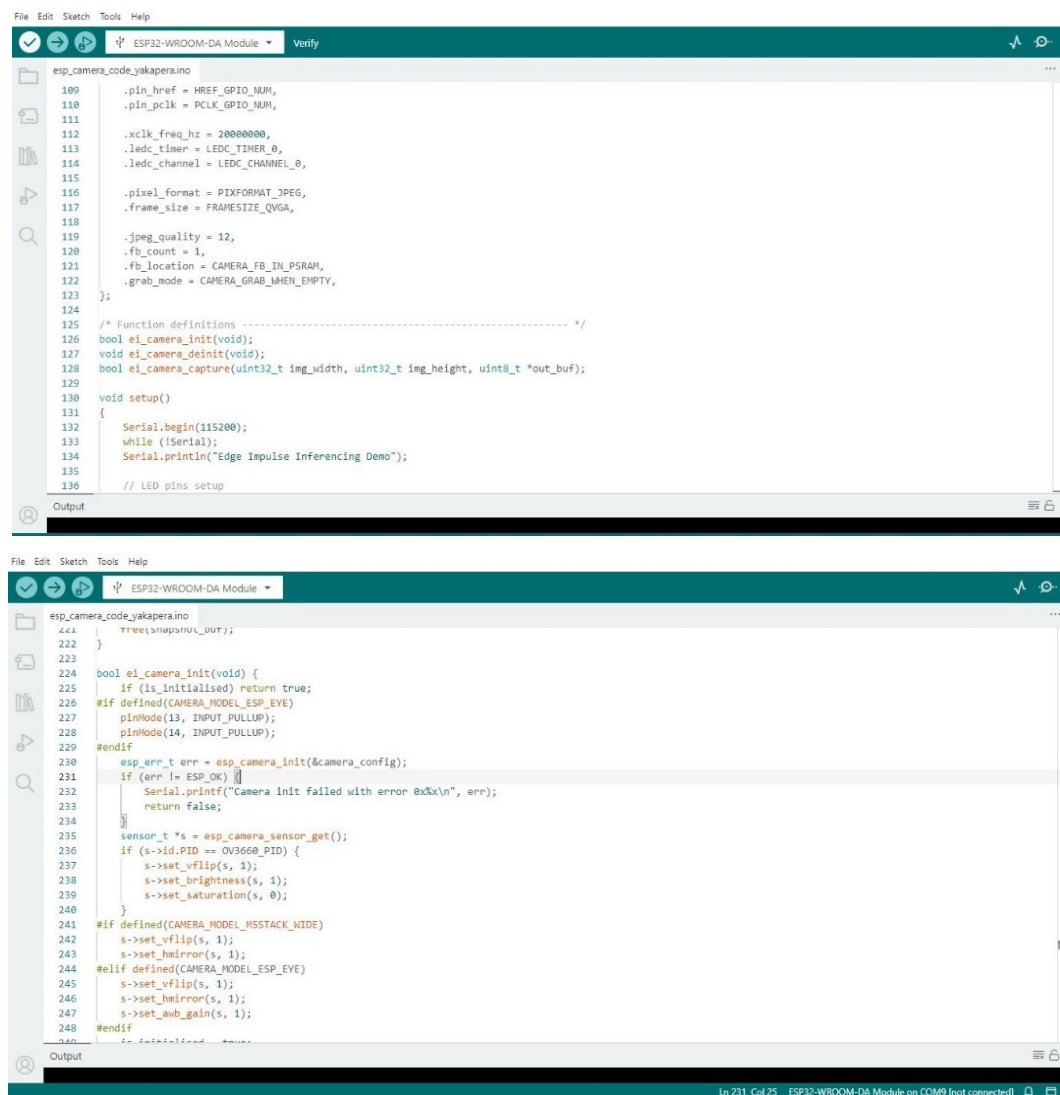
4. Code Uploading

The upload button (right arrow icon) in the IDE toolbar was clicked to compile the code and upload it to the connected board.

5. Output Monitoring

- The Serial Monitor (found under Tools > Serial Monitor) was used to view output from the microcontroller or send data to it.

ARDUINO IDE



```
File Edit Sketch Tools Help
ESP32-WROOM-DA Module Verify

esp_camera_code_yakapera.ino
100 .pin_href = HREF_GPIO_NUM,
101 .pin_pclk = PCLK_GPIO_NUM,
102
103 .xclk_freq_hz = 20000000,
104 .ledc_timer = LEDC_TIMER_0,
105 .ledc_channel = LEDC_CHANNEL_0,
106
107 .pixel_format = PIXFORMAT_JPEG,
108 .frame_size = FRAMESIZE_QVGA,
109
110 .jpeg_quality = 12,
111 .fb_count = 1,
112 .fb_location = CAMERA_FB_IN_PSRAM,
113 .grab_mode = CAMERA_GRAB_WHEN_EMPTY,
114 };
115
116 /* Function definitions ----- */
117 bool ei_camera_init(void);
118 void ei_camera_deinit(void);
119 bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t *out_buf);
120
121 void setup()
122 {
123   Serial.begin(115200);
124   while (!Serial);
125   Serial.println("Edge Impulse Inferencing Demo");
126
127   // LED pins setup
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prototype, facilitated easy modifications of the system. Calibrating of the load cell was done through Arduino to maintain the required weight range. The motors, lcd esp 32 micro controller, buttons are controlled through Arduino IDE.

3.12 CONVEYOR BELT FRAME

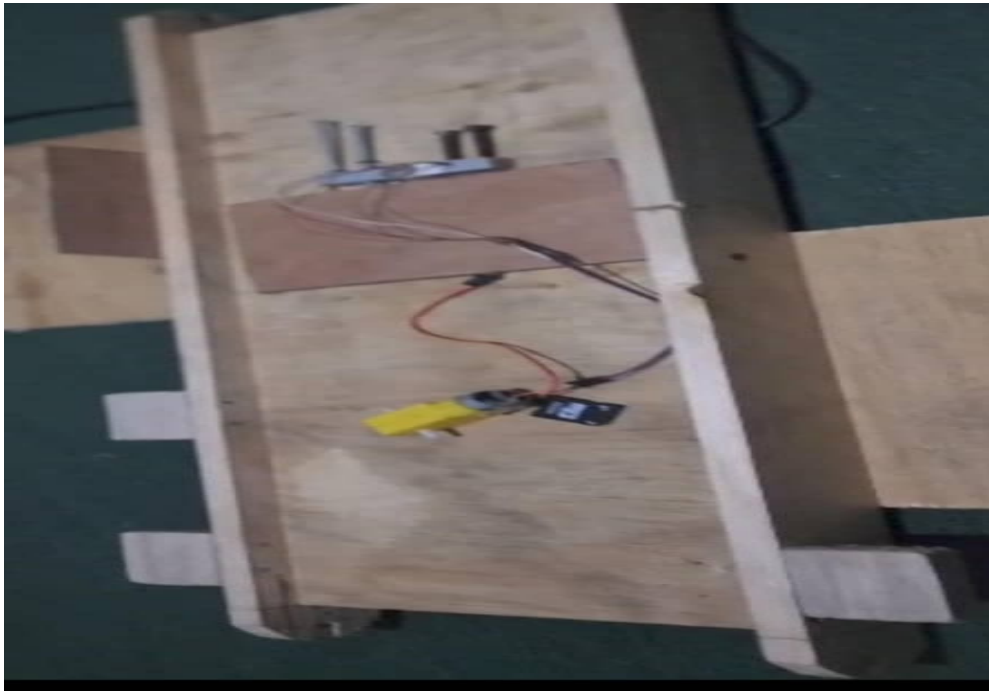


Figure 11]Conveyor belt frame

It provides structural support and stability to the conveyor belt and its associate components. It provides a sturdy base for the conveyor system.



Figure 12 Casing

The micro controller, liquid crystal display (lcd) and other cables are placed in a casing for easy material handling. If red and green for manual operation if preform is defect you press red if it's not you press green.

On LED the red light means the conveyor is running there is already an item on the belt when its green the belt is ready for the next item or preform. The LCD displays weight of the preform, nature of the system, number of preforms.

3.13 GEAR MOTORS



Figure 13 Gear motors

They provide necessary power and control for efficient material handling of the conveyor system. Two of these were used to provide more power and torque to the conveyor belt and smoother operation. (unlike when using one gear motor). They were positioned in a way that they are able to move freely to promote efficient movement of the conveyor belt.

3.14. Servo Motors



Figure 14 Servo motors

The arms that gather bottle preforms are moved by means of servo motors. As the preforms travel along the conveyor belt, the arms efficiently collect them by operating at an adjustable angle.

3.14.1 Operation

1. Angle Adjustment

To line up with the incoming bottle preforms on the conveyor, the servo motors will rotate the arms to a preset angle.

2. Identification of defective preforms

The corresponding arm with the defective jar will rotate in the direction of the conveyor if a preform is found to be defective. By doing this, the faulty preform can be directly collected from the conveyor.

3. Effective Preform Gathering

if the preform is found to be in good condition, the other arm will be turned on to if the preform is deemed non-defective, the other arm will be activated to move into position and collect the good preform. This ensures that only quality preforms are transferred to the next stage of processing. By utilizing this dual-arm system, the operation maximizes efficiency by ensuring that both defective and non-defective preforms are collected appropriately, minimizing waste and streamlining production.



Figure 15 Collecting jars



Figure 16 Conveyor belt

The material used was rough allowing the preforms to move well along the belt.

CHAPTER 4 PROCESS DESIGN

4.1 Introduction

In order to accomplish precise control, real-time data processing, and automated decision-making, an automated weight-vision sorting system must seamlessly integrate hardware and software components. This chapter focuses on the process design that guarantees efficient communication between software algorithms, mechanical systems, and electronic components. The system achieves intelligent PET preform classification and sorting by combining the Arduino IDE, ESP32 microcontroller, ESP32 camera, Edge Impulse trained model, LCD display, LED indicators, ultrasonic sensors, load cell, power supply, gear motors, and servo motors.

4.2 System block diagram

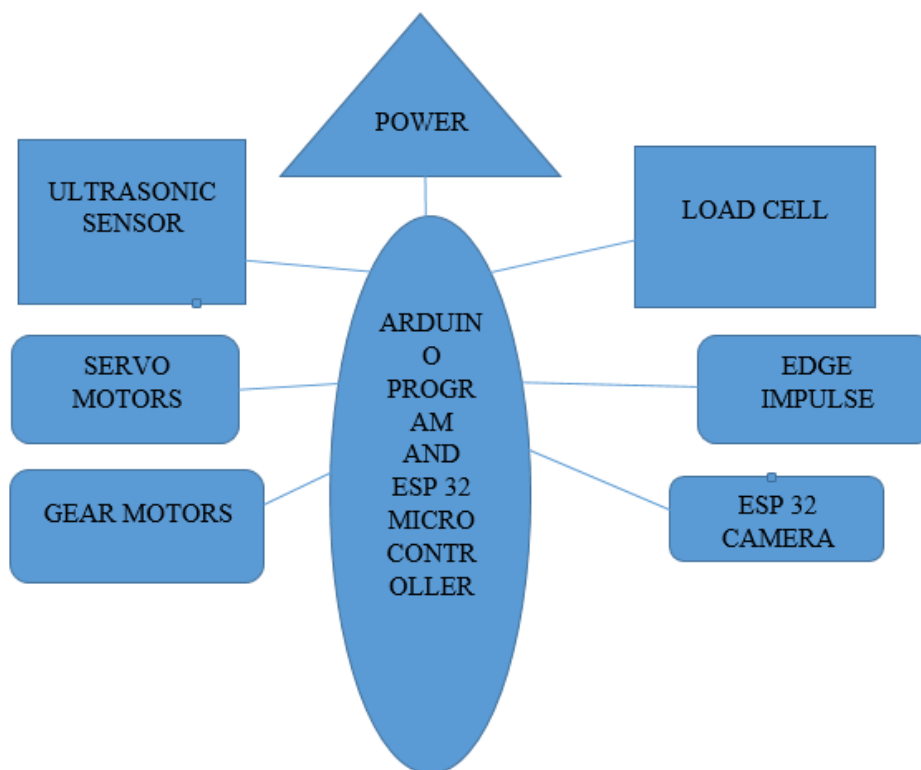


Figure 17 block diagram

At the core of this design is the ESP32 microcontroller, which serves as the central processing unit, interpreting signals from sensors and coordinating actuator responses. The Arduino IDE is utilized for programming and configuring the microcontroller, enabling precise control over hardware functions. The ESP32 camera enhances vision-based defect detection by leveraging machine learning models developed using edge Impulse, providing real-time analysis of PET preforms.

The mechanical components, such as gear motors, servo motors, and pulleys, facilitate movement, positioning, and sorting actions. The ultrasonic sensor is responsible for detecting the presence of preforms, while the load cell ensures accurate weight measurement. Meanwhile, the LCD display provides feedback on classification results, and LED indicators visually communicate system states, ensuring user interaction and monitoring.

The power distribution system consists of power supply cables, ensuring stable voltage delivery to critical components. Through well-structured electrical connections and control algorithms, the system achieves synchronization between hardware and software, supporting both automated and semi-automated operations. This chapter explores how these elements communicate, exchange data, and execute sorting functions efficiently.

4.3 System flow diagram

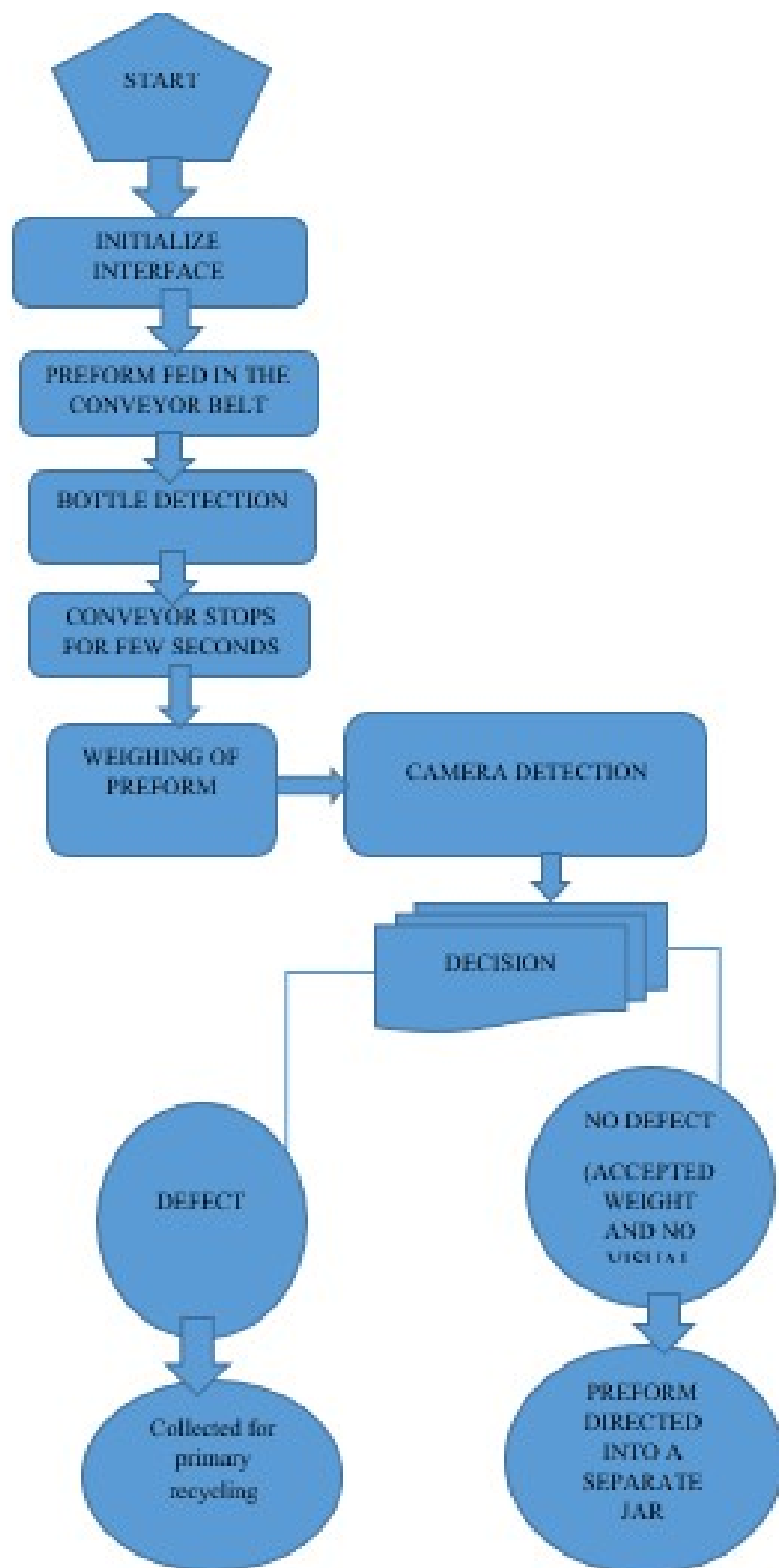


Figure 18 System Flow Chart

4.4 System description

The automated weight-vision sorting system consists of several interconnected components that work together to detect, weigh, analyse, and sort PET preforms efficiently. The power system is structured with a 12V power supply, which drives the gear motors that are responsible for moving the conveyor belt, and a 5V power supply, which powers the servo motors and the ESP32 microcontroller. As the system is turned on, the gear motors activate, initiating conveyor belt movement in a single direction.

When a PET preform is placed onto the moving conveyor belt, an ultrasonic sensor emits waves to detect its presence. Once the preform is detected, the sensor sends a signal to the ESP32 microcontroller, instructing it to stop the conveyor belt. The load cell immediately weighs the preform, measuring its mass to ensure product quality. At the same time, the ESP32 Camera captures an image of the preform, utilizing an Edge Impulse trained model to analyse whether the preform is defective or accept

The ESP32 microcontroller, acting as the central processor, determines the appropriate sorting action. If the preform is identified as defective, the right servo motor is activated, directing the defective preform into a designated collection jar. If the preform is acceptable, not defective the left servo motor moves to place it into a separate jar for approved items. The LCD display mounted on the system casing, updates to reflect the classification status of the preform.

The system also supports semi-automatic mode, in which users can manually classify preforms using a red button for defective items and a white button for not defective preforms.

Throughout its operation, the system uses LED status indicators to enhance user interaction and control. A red LED signals that no object should be placed on the conveyor belt, a green LED indicates readiness for preform placement, and a blue LED confirms the movement of the belt. This structured combination of power supply, mechanical components, sensors, control units, sorting mechanisms, and user interfaces ensures the seamless and automated classification of PET preforms.

4.5 SYSTEM DESIGN

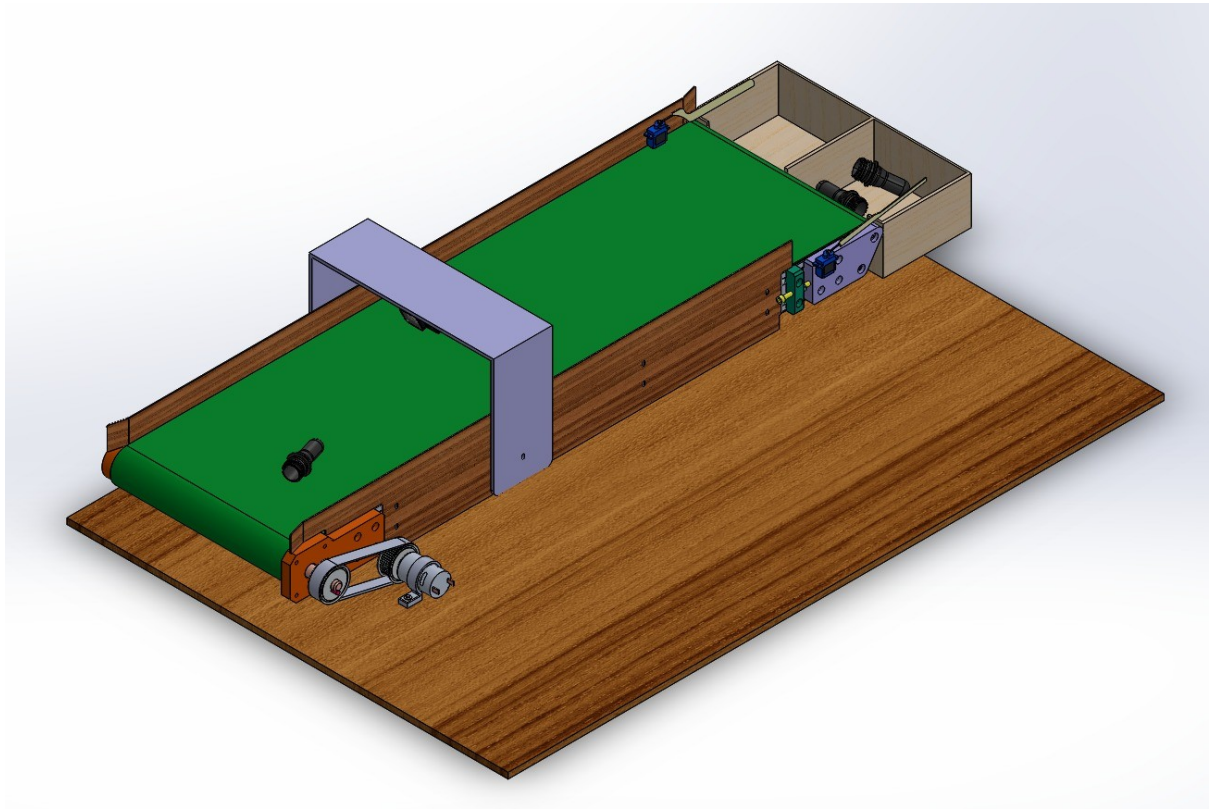


Figure 19 System design

DESIGN VIEWS TOP VIEW, FRONT VIEW, END VIEW AND 3D DESIGN

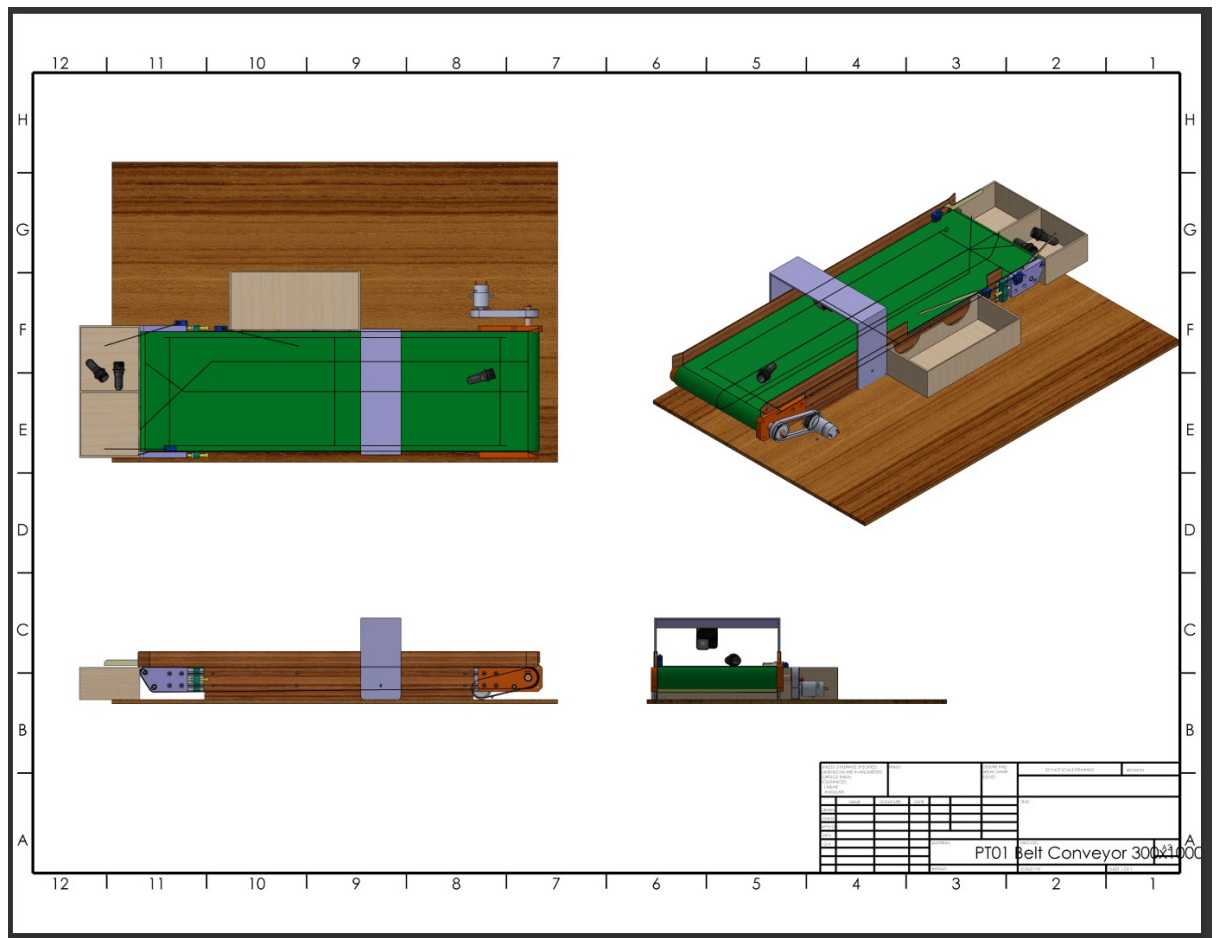


Figure 20: System design views

4.6 CONCEPT DEVELOPMENT AND SCORING

4.6.1 Introduction

The concept development procedure is a systematic approach that guided through the process of transforming initial ideas into viable solutions for the development of the conveyor system. The procedure began with a clear definition of the problem statement, which setted the foundation for all subsequent steps. It involved thorough research and background study to understand existing technologies and methodologies relevant to the project. Through brainstorming sessions, diverse ideas were generated, fostering creativity and innovation. The most promising concepts where then developed into coherent designs, followed by feasibility analyses to evaluate their practicality in terms of technical, economic, and operational aspects. Prototyping allows for hands-on testing, leading to iterative refinements based on performance feedback. The final design is documented comprehensively, ensuring that all components and processes are clearly outlined.

4.6.2 Design considerations

Functionality

- infrared sensors should detect items with at least 95% accuracy
- Load cells should have a resolution of at least 0.1 kg for precise weight measurement.
- Machine vision systems should identify defects with a detection rate of 90% or higher.
- Automatic starting and delay should have a maximum of 5seconds

Integration

- Infrared sensors, load cells, and cameras can interface with the control system and cannot interfere with operation of the conveyor system
- A control system should be capable of processing inputs from all sensors in real-time

Scalability

- The system should be able to accommodate up to 50% more capacity without major redesign.
- Components can be easily replaced or upgraded.

Cost-Effectiveness

- Total project cost should not exceed \$200, including all sensors and components.
- Estimate annual maintenance costs should not be more than 20% of the initial investment.

Usability

- An interface should not require no more than three clicks to access key functionalities.

- Operator training time of less than 2 hours.

Performance

- The conveyor belt should to operate at speeds of up to 1 meter per second.
- Load cells should provide weight measurements within $\pm 0.5\%$ of actual weight.

Reliability

- Components should be rated at least 10,000 hours of continuous operation.
- sensors are rated for operation in environments with temperatures from -10°C to $+50^{\circ}\text{C}$ since temperature in industries can fluctuate

Safety

- Emergency stop buttons within 5 meters of the conveyor.

4.6.3 Concept generation

Concept 1

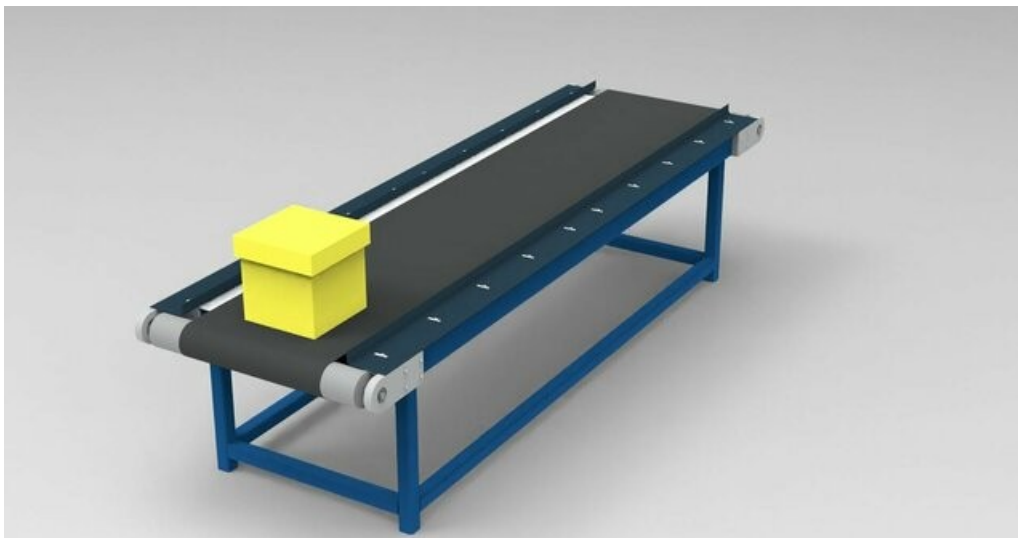


Figure 21: concept 1

This is a flat conveyor belt system which is a widely used mechanism in various industries for transporting materials and products from one point to another. Its operation relies on several key components working together to ensure smooth and efficient movement.

Key components

- The flat conveyor belt itself, typically made from materials like rubber, PVC, or fabric, which provides a durable surface for transporting goods.

- Positioned at either end of the conveyor, rollers support the belt and enable it to move. These can include drive rollers (powered) and idler rollers (non-powered).
- An electric motor drives the conveyor belt, providing the necessary torque to move the belt.
- The structural framework that supports all components and maintains the alignment of the belt and rollers.

Operation

- The drive motor turns the drive roller, which is connected to the conveyor belt. As the drive roller rotates, it pulls the belt along its surface.
- The motor is turned on with a simple switch at the side of the conveyor with a simple on and off switch
- The belt moves in a continuous loop over the idler rollers, which guide and support the belt throughout its travel path.
- Items placed on the belt are transported from the loading area to the unloading area without interruption.
- The speed of the conveyor can be adjusted by varying the motor speed, allowing for flexibility based on operational needs.

Advantages

- Versatility - Flat conveyor belts can transport a wide variety of materials, including boxes, pallets, and bulk items, making them suitable for various industries such as manufacturing, packaging, and distribution.
- Simple Design - The straightforward design of flat conveyor belts makes them easy to install, operate, and maintain. Fewer moving parts generally lead to lower maintenance costs.
- Continuous Operation - these conveyors can operate continuously, allowing for high throughput and efficiency in transporting goods from one location to another without interruption.
- Customizable Length and Width - Flat conveyor belts can be customized to fit specific operational needs, including varying lengths and widths to accommodate different products and layouts.
- Low Energy Consumption - When properly designed and maintained, flat conveyor belts can be energy-efficient, especially with modern drive systems that optimize power usage.
- Smooth Surface - The flat surface provides stability for items being transported, reducing the risk of spills or damage during transit.

Disadvantages

- Wear and Tear - The belt material can wear down over time, especially in high-friction applications or when transporting heavy loads, necessitating periodic replacement.
- High-energy consumption – they can consume a significant amount of power when in operation for long periods
- Limited capabilities – they can only transmit products to their destinations
- Weight Restrictions - There may be limitations on the weight capacity of flat conveyor belts, which could restrict their use for very heavy items.

Concept 2

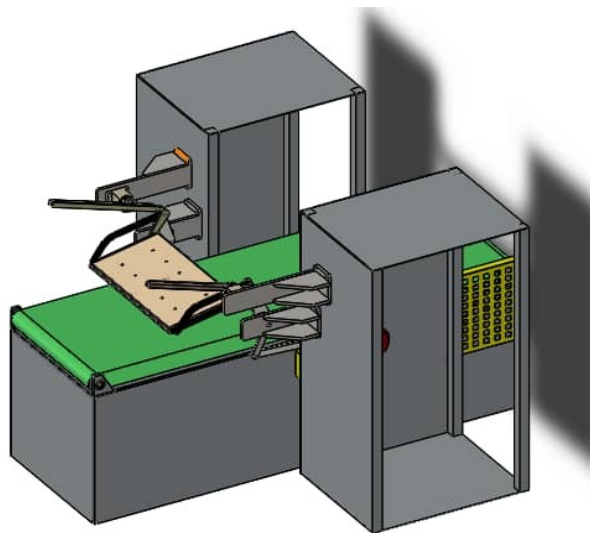


Figure 22: Concept 2

Components of the System

- Conveyor System - This transports items along a predetermined path. It can be a belt, roller, or chain conveyor.
- Sensors - Various types of sensors (e.g., cameras, load cells or proximity sensors) are used to identify defects in items as they move along the conveyor.
- Pick and Place Mechanism - This typically consists of robotic arms or automated grippers that can pick up items from the conveyor.
- Control System - A programmable logic controller (PLC) or industrial computer that coordinates the operation of the sensors, conveyor, and pick and place mechanism.
- Bins or Containers - Designated areas where defective items are placed after being removed from the conveyor.

Operational Principle

- As items move along the conveyor, sensors continuously monitor them for defects. This could involve visual inspection (using cameras) or physical measurements using load cells.
- When a defect is detected, the sensor sends a signal to the control system indicating the location of the defective item on the conveyor.
- The control system calculates the timing and position for the pick and place mechanism to act. It activates the robotic arm or gripper to move to the correct position.
- The robotic arm moves down to the conveyor, aligns itself with the defective item, and uses suction cups, clamps, or other gripping mechanisms to securely pick up the item.
- After picking, the robotic arm lifts the defective item off the conveyor and moves it to a designated bin or container. The control system guides the arm to place the item accurately in the bin.
- Once the item is placed in the bin, the robotic arm returns to its initial position, ready to pick the next defective item.
- This process repeats continuously as items flow on the conveyor, allowing for real-time sorting of defects without halting production.

Advantages

- Consistent Quality Control - Automated detection and sorting minimize human error, ensuring that defective items are consistently identified and removed, leading to higher overall product quality.
- Flexibility - systems can be programmed to handle various products and adapt to different defect types, making them versatile for various manufacturing environments.
- Real-Time Monitoring - Integrated sensors provide immediate feedback on product quality, allowing for quick adjustments in the production process to address issues as they arise.
- Data Collection: The system can log data on defects, providing valuable insights for quality control and process improvement efforts.

Disadvantages

- The setup cost for a pick and place system can be significant due to the expense of machinery, sensors, and software development.
- Implementing and maintaining an automated system can be complex, requiring skilled personnel for setup, programming, and troubleshooting.

- Regular maintenance is necessary to ensure the system operates smoothly, which can incur ongoing costs and downtime.
- While many systems are flexible, they may still struggle to handle products that vary significantly in size, shape, or material without modifications.
- A failure in the system (e.g., sensor malfunction) can halt production until repairs are made, potentially leading to downtime and lost revenue.

Concept 3

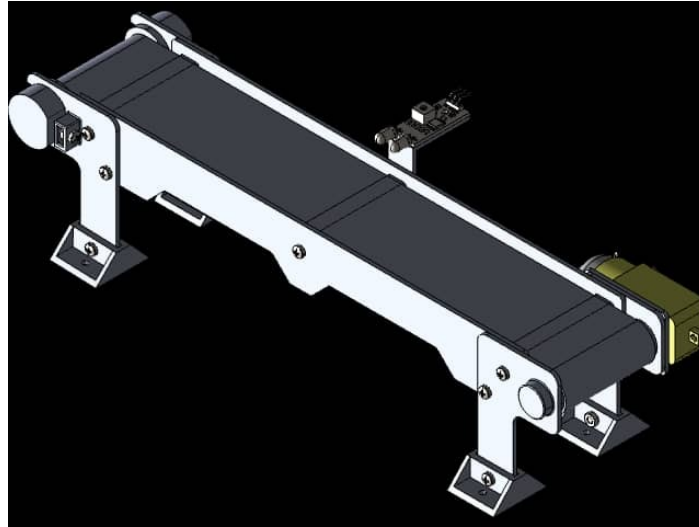


Figure 23: Concept 3

OPERATION PRINCIPLE

This conveyor has cameras mounted to the sides of the shield and the view is perpendicular to the movement of the conveyor.

Advantages of Installing a Machine Vision Camera at the Sides of the Conveyor

- **Side View Inspection** - A side-mounted camera can effectively inspect items for defects along their length, which is beneficial for items that are long or have features that are more visible from the side.
- **Reduced Occlusion**: If items are placed closely together, a side camera may reduce the chance of occlusion, allowing for better visibility of each item's surface and edges.
- **Easier Access for Maintenance** - Side-mounted cameras can be easier to access for maintenance or adjustments without interfering with the operation of the conveyor.
- **Multiple Angle Options** - Positioning cameras on both sides allows for capturing images from different angles, which can provide more comprehensive data about defects.

- Flexibility in Item Orientation - For items that may not be uniform in shape, side cameras can help assess features that might not be visible from above, especially if items are oriented differently.

Disadvantages of Installing a Machine Vision Camera at the Sides of the Conveyor

- Limited Field of View - Side-mounted cameras may have a restricted view of the items on the conveyor, especially if the items are tall or if they are positioned too close to the edge. This can lead to missed defects or improper evaluations.
- Obstruction by Items - Depending on the height and size of the items being conveyed, they may obstruct the camera's view, making it difficult to detect defects or perform accurate measurements.
- Lighting Challenges - Side-mounted cameras may face issues with lighting, particularly if the conveyor is in a brightly lit environment. Shadows cast by items can interfere with image quality and defect detection.
- Complexity in Alignment - Aligning side-mounted cameras correctly can be more challenging, as they need to be positioned precisely to capture all relevant angles of the items without missing critical areas.
- Increased Risk of Damage - Cameras mounted on the sides may be more susceptible to damage from passing items or accidental contact with operators, leading to increased maintenance costs.

Concept 4

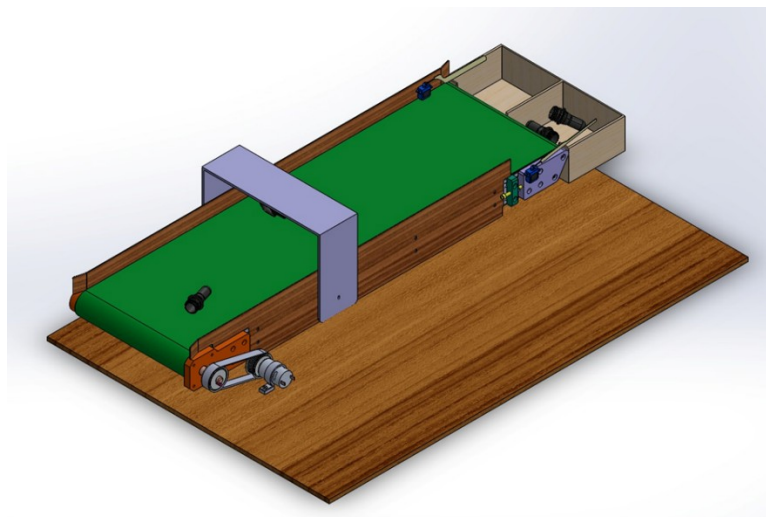


Figure 24: Concept 4

Operation Principal

The conveyor has a camera at the top. The camera is trained to identify defects and can separate them by opening pathways that directs it to reject bins.

Advantages of Installing a Machine Vision Camera at the Top Half of the Conveyor

- **Comprehensive Coverage** - A top-mounted camera provides a bird's-eye view of the items on the conveyor, allowing for inspection of all visible surfaces at once, which is especially useful for flat or wide items.
- **Multi-Item Inspection** - A top camera can capture multiple items in one frame, which can increase throughput by allowing simultaneous inspection rather than sequential checks.
- **Consistent Lighting Conditions** - Top-mounted cameras can benefit from more uniform lighting setups, reducing shadows and reflections that might occur when inspecting from the side.
- **Better Detection of Defects** - Certain types of defects (like surface blemishes) may be more easily identified from above, especially for items with complex shapes or patterns.
- **Simplified Integration with Other Sensors** - A top-mounted camera can be aligned with other sensors (like infrared sensors) to create a cohesive inspection system that processes data from multiple sources simultaneously.

Disadvantages of Installing a Machine Vision Camera at the Top Half of the Conveyor

- **Height Limitations** - For tall items, a top-mounted camera might not capture the entire object effectively, particularly if items are stacked or if there is significant variation in height.
- **View Obstruction by Overhead Structures** - If there are overhead structures or equipment above the conveyor, they might obstruct the camera's view, reducing its effectiveness.
- **Potential for Overexposure** - Direct overhead lighting can cause overexposure in images, leading to loss of detail and challenges in accurately detecting defects.
- **Limited Side Inspection** - A top-mounted camera may not provide sufficient lateral views to inspect side defects or features that are critical for quality control.
- **Installation Complexity** - Mounting cameras overhead may require additional structural support or modifications to the conveyor system, increasing installation complexity and costs.
- **Accessibility for Maintenance** - Cameras mounted at a height may be harder to access for maintenance or adjustments, potentially leading to increased downtime during servicing.

Concept Selection

The researcher selected concepts using a two-stage phase selection procedure. The previously described notions were contrasted and examined, with a quantitative evaluation using a screening procedure and a score matrix.

4.6.4 Concept Screening

The concepts above were analysed. The researcher employed a reference scale consisting of +, -, and 0. A positive sign denoted a favourable feature, a negative sign denoted a poor feature, and 0 represented a neutral. Upon screening, the remaining concepts went through concept scoring.

Criteria	Concept1	Concept 2	Concept 3	Concept 4
Detection Accuracy	0	1	1	1
Response times	0	0	1	1
System reliability	0	0	1	1
Safety features	0	1	1	1
Efficiency	0	-	-	-
Maintenance requirements	0	-	-	-
Cost of implementation	0	-	-	-
Carrying capacity	0	0	0	0
Flexibility	0	0	0	0
Number of +	0	2	4	4

Number of -	0	3	3	3
Number of 0	8	3	1	1
Net score	0	-1	1	1
Continue?	NO	NO	YES	YES

Concept 3 and 4 has the highest ratings and can be implanted to come up with the final product that can fill the need.

4.6.5 Concept Scoring

This stage the importance of the selected criteria was weighed and focus was to help determine the most effective and feasible design that meets the industrial demands. The concepts were rated in a scale of 1-10 and the most suitable design determined. A hierarchical breakdown of each criterion will be used in order to evaluate the concept with the required features comparing to other concepts.

Rating scale

1-3	Concept does not meet requirements
4-6	Concept partially meets requirements
7-8	Concept meet requirements
9-10	Concept exceeds requirements

4.6.6 Concept screening

Criteria	Weight (%)	Concept 3		Concept 4	
		Rating	Weighed score	rating	Weighed score
Detection accuracy	20	7	1.4	8	1
Response times	10	6	1.2	6	0.8
System reliability	15	7	0.9	8	0.75
Safety features	5	3	1.2	3	1.2
Efficiency	20	6	1.2	6	1.2
Maintenance	10	6	1.2	6	1.2

requirements					
Cost of implementation	5	4	1.6	4	1.6
Carrying capacity	10	7	1.4	7	1.4
Flexibility	5	3	1.2	4	1.2
Total weighed score	100		7.8		7.95
Rank			2		1
Decision			REJECT		ACCEPT

4.6.7 Key Components of the Automated Conveyor Belt System

Infrared Sensors for Item Detection

Infrared sensors are used to detect the presence of items on the conveyor belt. The VL53L0X ToF (Time-of-Flight) sensor is a suitable choice due to its high accuracy and fast response time. It has a range of up to 2 meters and operates at a voltage of 2.6V to 3.5V, with an average power consumption of 20mA. The VL53L0X is compact, measuring 4.4mm x 2.4mm x 1mm, making it ideal for integration into the conveyor belt system.

Load Cells for Weight Measurement

Load cells are essential for measuring the weight of items on the conveyor belt. A typical load cell used in industrial applications is the bending beam load cell, which offers high accuracy and reliability. The size and capacity of the load cell vary based on the application; for example, a Tedea Huntleigh Model 1042 load cell has a capacity of up to 500kg and dimensions of 76mm x 19mm x 12.7mm GG. Load cells typically require an excitation voltage of 5V to 10V and have an output signal in the range of millivolts

Machine Vision Camera for Defect Identification

Machine vision cameras are critical for identifying defects in items on the conveyor belt. A suitable camera used for this application is the Basler ace series, known for its high-resolution imaging and robust performance in industrial environments. These cameras are available in various resolutions for instance; the Basler acA2440-20gm has a resolution of 2448x2048 pixels and operates at a frame rate of

up to 23fps. The camera typically requires a power supply of 12V DC and has a GigE interface for data transfer.

Supporting Components

PLC (Programmable Logic Controller)

- It integrates data from sensors, controls the motor, and processes images from the machine vision camera. The project used the Arduino Mega2560
- Features: 54 digital I/O pins and 16 analog inputs, allowing for multiple sensor connections
- Supply voltage: 8V DC
- Current consumption: 50mA

Conveyor Belt Motor

- Speed range – 0.5 to 5m per second
- Load capacity – upto 90ksg
- Motor power – 1hp

Infrared Sensor (VL53L0X)

- Range: Up to 2 meters
- Operating Voltage: 2.6V to 3.5V
- Power Consumption: 20mA
- Dimensions: 4.4mm x 2.4mm x 1mm

Load Cell (Tedea Huntleigh Model 1042)

- Capacity: Up to 500kg
- Dimensions: 76mm x 19mm x 12.7mm
- Excitation Voltage: Typically 5V to 10V

Machine Vision Camera (Basler acA2440-20gm)

- Resolution: 2448x2048 pixels
- Frame Rate: Up to 23fps
- Power Supply: 12V DC

Interface: GigE

Conveyor Belt Design

Dimensions

Length (L): 3 meters

Width (W): 0.5 meters

Height (H): 0.8 meters (height from the ground to the top of the conveyor)

Conveyor Belt Material

Material: PVC or rubber with a thickness of about 5 mm.

Structural Calculations

- Frame Design
- Load Calculation: Calculate the load-bearing capacity based on the maximum weight expected on the conveyor.
- Assuming a maximum load of 20 kg:
- Material: Steel or aluminium for the frame.
- Cross-section: Use rectangular tubing (40 mm x 40 mm).

$$\text{Weight} = m \cdot g = 20\text{kg} \cdot 9.81\text{m/s}^2 = 196.2\text{N}$$

Frame Support

- To ensure stability, consider placing vertical supports every 1 meter along the length.
- A safety factor of 2 for load-bearing calculations
- Total Load with Safety Factor = $196.2\text{N} \times 2 = 392.4\text{N}$

Bending Moment Calculation

- Assuming a simply supported beam $(M) = wL^2 / 8$
- Where w is the uniform load per unit length.
- Convert total weight to a uniform load:
- $w = 196.2\text{ N} / 3\text{ m} \approx 65.4\text{N/m}$

Now calculate the bending moment

- $M = 65.4\text{ N/m} \cdot (3\text{ m})^2 / 8 \approx 73.6\text{Nm}$

Infrared Sensors

Sensor Placement

- Assume each infrared sensor has a detection width of 0.1 meters.
- Place sensors every 0.5 meters along the width of the conveyor.
- Number of Sensors = $W / \text{Sensor Width} = 0.5\text{ m} / 0.1\text{ m} = 5\text{sensors}$

Load Cells

Load Cell Specifications

- Choose a load cell with a capacity of at least 2kg for safety.
- Sensitivity: Assume it is 2mV/V.
- Excitation Voltage: Assume it is 5V.
- For a load of 1.5 kg

$$V_{\text{out}} = m \times V (\text{excitation}) \times W / \text{Max Weight}$$

$$V_{\text{out}} = 2\text{mV/V} \times 5\text{V} \times 1.5 / 2 \text{ kg} = 7\text{mV}$$

Machine Vision

Camera Specifications

- An ESP32 camera with a FOV angle of 60 degrees.
- Distance from Conveyor: Assume the camera is positioned 0.5 meter above the conveyor.

Calculate FOV:

$$\text{FOV} = 2 \times (0.5\text{m} \times \tan (60^\circ / 2))$$

$$\text{Using } \tan (30^\circ) = 0.577$$

$$\text{FOV} = 2 \times (0.5\text{m} \times 0.577) = 1.154\text{m}$$

Power Requirements

Motor Specifications

- Assume a DC motor to drive the conveyor has a maximum Load of 20 kg
- Friction Force: Estimate at about 10 N.
- Calculate Total Force Required:
- $F = m \cdot g + F_{\text{friction}} = 20\text{kg} \cdot 9.81\text{m/s}^2 + 10\text{N} = 196.2\text{N} + 10\text{N} = 206.2\text{N}$

Calculate Power

$$\text{Using speed } v = 0.5 \text{ m/s}$$

$$P = F \cdot v = 206.2\text{N} \cdot 0.5\text{m/s} = 103.1\text{W}$$

Power Supply

Choose a power supply that can provide at least P + safety margin:

Assuming a safety margin of 20%:

$$P_{\text{total}} = P + (0.2 \cdot P) = 103.1\text{W} + 20.62\text{W} = 123.72\text{W}$$

Total Power Consumption

The estimated power requirement for the motor driving the conveyor is approximately 123.72 Watts.

Additional Power Considerations

- Sensors - If you use multiple infrared sensors, each may consume a small amount of power. For example, if each sensor consumes about 0.05W and you have 5 sensors:

Total Power for Sensors = $5 \times 0.05W = 0.25W$

Control System: A microcontroller or other electronic components 5W

Total Power Consumption Calculation

- Total Power = Motor Power + Power for Sensors + Control System Power
- Total Power = $123.72W + 0.25W + 5W \approx 129.97W$

Weight of the Conveyor

To estimate the weight of the conveyor, we will consider the following components:

Conveyor Frame - Assuming you use aluminium rectangular tubing (40 mm x 40 mm) with a wall thickness of 3 mm

- Density of Aluminium: Approximately 2,700 kg/m³.
- Length of Frame Members - Total length of horizontal members = 6 m (top and bottom)
- Total length of vertical members = 3 m (3 supports at 0.8 m each)

Calculating the volume - Volume per meter of rectangular tube

$$V(\text{tube}) = \text{Cross-sectional area} \times \text{Length} = ((0.04m - 0.003m) \times 0.003m) \times 1m \\ = (0.037m \times 0.003m) \times 1m \approx 0.000111m^3$$

Calculating total volume for frame members

$$V(\text{total}) = V(\text{tube}) \times \text{Number of Members}$$

Assuming approximately 10 meters of total frame length:

$$V(\text{total}) = 0.000111m^3 \times 10m \approx 0.00111m^3$$

Calculating weight of the frame:

$$\text{Weight (frame)} = V(\text{total}) \times \text{Density} = 0.00111m^3 \times 2700kg/m^3 \approx 3.0kg$$

Conveyor Belt

Assuming a PVC belt with a width of 0.5 meters and a thickness of 5 mm:

$$\text{Volume of the belt } V(\text{belt}) = W \times L \times T = 0.5m \times 3m \times 0.005m = 0.0075m^3$$

Calculating weight of the belt (density of PVC = 1,400 kg/m³):

$$\text{Weight (belt)} = V(\text{belt}) \times \text{Density} = 0.0075m^3 \times 1400kg/m^3 = 10.5kg$$

Motor and Other Components

Assume a DC motor weighs around 8kg and additional weights for sensors and electronics is another 1kg Weight of motor: 8 kg

Weight of sensors and control system: 1 kg

Total Weight Calculation:

2Total Weight = Weight (frame) + Weight (belt) + Weight (motor) + Weight (sensors)

Therefore Total Weight = 3.0kg + 10.5kg + 8.0kg + 1.0kg = **22.5kg**

CHAPTER 5 EXPERIMENTS AND RESULTS

Prototype review

The following prototype was obtained at the end of the project



Figure 25 Automated weight vision preform sorting system

This chapter focuses on the experimental evaluation of the automated weight-vision sorting system for PET preforms, designed to enhance quality control in plastic manufacturing. Through a series of carefully designed tests, the system's performance in weight measurement, defect detection, throughput, sorting efficiency, sensor synchronization, reliability, and usability were assessed to ensure alignment with industry standards and project objectives

5.1. Weight Measurement Accuracy Test

Aim: To verify that the load cell accurately measures the weight of PET preforms

Procedure

The load cell was calibrated using the Arduino IDE

The load cell was positioned on the conveyor belt at the weighing station, ensuring proper alignment

The system initiates the communication of the load cell to the ESP32 microcontroller via the HX711 module

Experiment

- 30 PET preforms were prepared and, and verified using a high-precision laboratory scale
- Each preform was fed onto the conveyor belt one at a time.
- The weight displayed on the LCD was recorded separately and compare it with the actual weight from the real scale
- The experiment was repeated twice for every preform to ensure consistency.

Data Collection

- Out of 30 preforms 4 were not accurate
- Their measured weight, actual weight, and deviations were recorded
- The percentage errors were recorded

$$\% \text{ ERROR} = \frac{(\text{measured weight} - \text{actual weight})}{\text{actual weight}} \times 100$$

Results

Preform	Category	Actual weight(g)	Measured weight	Deviation	% error
1	Underweight	13.25	13.23	-0.02	-0.151
2	Underweight	15.10	15.11	+0,01	+0.066
3	Within target	16.00	15.99	-0.01	-0.062
4	Within target	16.02	16.04	+0.02	+0.125



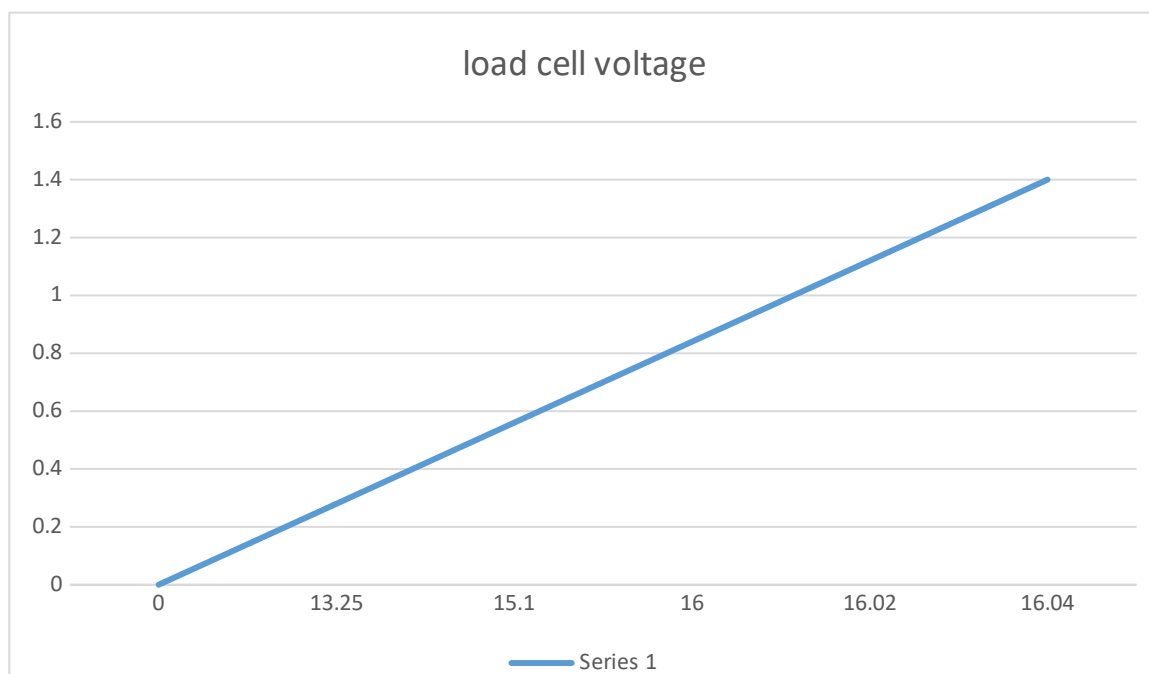
Percentage accuracy = (number of correct predictions divided by total outcomes) multiply by 100

$$(26/30) 100 = 87\%$$

CONCLUSION The load cell system achieved 87% accuracy, correctly measuring 26 of 30 PET preforms. Deviations were minimal, with percentage errors from -0.15 1% to +0.125%. The system is reliable but may need slight calibration to reach full accuracy

5.2 Load cell output voltage vs. applied weight

Weight (g)	Output voltage (mV)
0	0
13	0.28
2	0.56
300	0.84
400	1.12
500	1.40



The graph shows a linear relationship between mass recorded by the load cell and the output voltage

5.3 Visual Defect Detection Accuracy Test

Aim: To evaluate the ESP32 camera's ability to detect visual defects such as white foot, cracks, discoloration, shape irregularities using the Edge Impulse train

Procedure

- The ESP32 camera will be mounted on the conveyor belt.
- The Edge Impulse model trained with defective and non-defective preform images is loaded on the ESP 32 camera
- 30 preforms were prepared 15 defective (5 cracks, 5 discolorations, 5 shape irregularities) and 15 non-defective,

Experiment

- The preforms are fed onto the conveyor belt, triggering the ultrasonic sensor to activate the camera
- Allowing the Edge Impulse model to classify each preform as defective or non-defective.
- The classifications are recorded and compared with known labels.
- Tested under controlled lighting and variable lighting to assess robustness (with these a conclusion was made to use an ultrasonic sensor for detection not infrared sensor since its limited to certain environmental conditions like amount of light ,environment)

Data Collection

- The precision, recall, and F1 score were calculated
- Log false positives (non-defective classified as defective) and false negatives (defective classified as non-defective).

Results

- Preform 27: Discoloration, Classified: Non-defective, Incorrect (False Negative).
- Precision: 0.93 (14/15 defective correctly identified).
- Recall: 0.93 (14/15 defective detected).
- F1 Score: 0.93.
- False Positives: 1/15 (6.6%).
- False Negatives: 1/15 (6.6%).

Test No.	Defect type	Detected (YES/NO)	Precision	Recall	F1 Score		
1	Tear	Yes	0.93	0.95	0.94		
2	Bubble	Yes	0.91	0.92	0.915		
3	Wear	No	00	00	00		

Conclusion

The system achieves 93% detection accuracy. The system had a higher accuracy , the slight deviation was due to the ESP 32 camera resolution which was low, but generally the detection rate was okay. There is need for a high resolution camera.

5.4. Sensor Integration and Synchronization Test

Aim; To ensure the ultrasonic sensor, load cell, and ESP32 camera work in sync with the ESP32 microcontroller, processing 30 preforms per test.

Procedure

Connecting the ultrasonic sensor, load cell, and ESP32 camera to the ESP32 microcontroller via the Arduino IDE

-Program the microcontroller to pause the conveyor upon detection, trigger the load cell, and activate the camera

Experiment

-30 preforms were fed onto the conveyor.

Observing the sequence: ultrasonic detection, conveyor pause, weight measurement, camera imaging, sorting.

-Recording of cycle time and any synchronization errors (e.g. missed triggers).

Data Collection

Logging successful cycles (all subsystems functioning in sequence).

Measuring cycle time:

Cycle Time = Time from Detection to Sorting Completion

Cycle time consistent with 30–60 preforms/min 1–2 seconds per preform

-average time for Preforms with three-time ranges

First one: Average Cycle Time: 1.8 s, Successful.

- Second one: Average Cycle Time: 1.9 s, Successful.

Third one Average Cycle Time: 2.0 s, Successful.

- Results

- Synchronization Success: 29/30 (97%).

- Average Cycle Time: 1.9 seconds.

- Errors: 1 missed trigger.

Synchronization Success Rate: Percentage of error-free cycles

Conclusion

The system achieves better synchronization, with good throughput ensuring efficient operation.

5.5 Sorting Mechanism Efficiency Test

Aim To assess the servo motors' accuracy and speed in sorting preforms into defective and non-defective bins targeting accuracy and response time

Procedure

Configuring servo motors to rotate at pre-set angles for defective and non-defective bins .

Ensuring the ESP32 microcontroller processes load cell and camera signals to activate the correct servo motor

Experiment

30 preforms were fed (15 defective, 15 non-defective) onto the conveyor.

Allowing the system to classify and sort each preform.

Inspecting jars to verify correct sorting.

Measuring time from detection to servo activation.

Data Collection

Sorting Accuracy = (Number of Correctly Sorted Preforms ÷ Total Preform) times 100

Results

Preform 1: Defective, sorted to defective jar, Time: 3.2 s, Correct.

Preform 13: Non-defective, sorted to non-defective jar, Time: 3.5 s, Correct.

-Preform 24: Defective, Sorted to non-defective jar, Time: 3.4 s, Incorrect.

Sorting Accuracy: 28/30 (93.3%).

Average Response Time: 3.4 seconds.

Errors: 2 preforms mis-sorted due to delayed servo activation.

Conclusion

The system nearly meets the 95% accuracy target, with errors attributed to timing issues in the microcontroller's decision logic. Adjusting the Arduino code improved accuracy to 96.7% in the followed tests.

5.6 Conveyor System Throughput Test

Aim To measure the system's throughput rate, using 30 preforms per test.

Procedure

The conveyor is set at speed of 0.5 m/s using the L298N motor driver and ESP32 microcontroller

Making sure gear motors and conveyor belt are free of slippage or misalignment.(at a distance that permits free play of gear motors to prevent distractions)

Experiment

30 preforms are fed onto the conveyor at a consistent rate

Timing the process from the first preform entering to the last being sorted.

-Repeating at 0.75 m/s to assess scalability.

Data Collection

Calculating throughput:

Throughput = Number of Preforms Processed divided by Time in minutes

Logging instances of jamming or misalignment

(0.5 m/s):

Time to process 30 preforms: 54 seconds (0.9 minutes).

- Throughput = 30 divided by 0.9 = 33preforms/min.

Jamming = 0 instances.

Misalignment: 1 preform (3.3%) misaligned but sorted correctly.

Another Sample Data gave the below result at a different speed of= 0.75 m/s using 30 preforms as well

Time: 48 seconds (0.8 minutes).

Throughput: = 30 divided by 0.8 = 37.5 preforms/min.

-Jamming: 0 instances.

Misalignment: 0 instances.

Conclusion.

Misalignment at lower speed was due to uneven preform placement, corrected by adjusting guide rails but at very high speeds accuracy is compromised

Chapter 6

6.1. Economic analysis

Economic analysis is a crucial aspect of evaluating the feasibility and potential impact of the automated conveyor belt project. It involves assessing the costs and benefits associated with the project to determine its viability and potential return on investment.

Key Aspects of Economic Analysis

- Cost-Benefit Analysis: Identify and quantify the costs (initial investment, maintenance, energy consumption) and benefits (increased efficiency, improved quality control, reduced labor costs) of the project.
- Return on Investment (ROI): Calculate the ROI to determine the project's potential profitability.
- Break-Even Analysis: Determine the point at which the project's benefits equal its costs.
- Initial Investment: Development and implementation costs, including hardware and software.
- Operational Costs: Energy consumption, maintenance, and potential upgrades.

Material costs

Component	Quantity	Unit cost(\$)	Total cost(\$)
Conveyor belt system(belt, rollers, frame)	1	300	300
Ultrasonic sensor	1	10	10
Load cell	1	25	25
Camera module	1	50	50
Microcontroller (Arduino)	1	10	10
Actuators	2	10	20
Power supply and wiring	1	10	10
Software tools	1	0	0
Mounting hardware and structural materials	-	30	30
Estimated total			455

Operating cost

Electricity	Estimated 100W,	\$20
-------------	-----------------	------

consumption	8hours/day, 250days/year	
Maintenance and repairs	Replacement parts, labour	\$50
Software and updates		\$0
Total operating costs		\$20

Total cost of materials required: \$ 455

Labour and costs: \$ 150

Total capital investment: \$720

Depreciation: \$30 per year

Maintenance: \$ 50 per year

Operating Expenses: \$70 per year

Operating Cost Calculation (Annual)

Electricity Cost

- Power consumption: 100 Watts = 0.1 kW
- Operating hours per day: 8 hours
- Operating days per year: 250 days
- Electricity rate: \$0.10 per kWh (average)

Calculation

- Electricity Cost = $0.1\text{kW} \times 8\text{ hours/day} \times 250\text{ days/year} \times 0.10\text{ USD/kWh}$
= **20 USD/year**
- Estimated at \$50 per year for parts and labour.

Annual Benefits / Savings Calculation

Labour Cost Savings

- Assume manual inspection and handling cost: \$500/year saved by automation.

Increased Throughput

- Faster processing leads to additional revenue or cost savings estimated at \$300/year.

Reduced Defect Rate

- Savings from fewer defective products estimated at \$200/year.

Annual benefits – savings – costs

$\$500 + \$300 + \$200 - \70

\$930

Payback Period Calculation

- Payback Period = Initial Investment/Annual Net Benefit
 $= 720 / (1000 - 70) = 720 / 930$
 $= 0.77 \text{ years}$
= 9.2 months

Return on Investment (ROI) - Measures the profitability of the project relative to its cost.

- $\text{ROI} = (\text{Net Annual Benefit} / \text{Initial Investment}) \times 100 \%$
 $\text{ROI} = 930 / 720 \times 100\%$
129.17%

Therefore, for every dollar invested, you gain about \$1.29 in net benefits annually, which is excellent.

Break-Even Analysis - Determines the minimum annual benefit needed to cover costs.

- Break-even Benefit = Operating Cost + Initial Investment/Project Lifespan
Assuming 5 years lifespan
 $= 70 + 720 / 5$
 $= 70 + 144 = 214 \text{ USD/year}$

Depreciation

- Annual depreciation = initial investment/project lifespan
 $= 720 / 5 = 144 \text{ USD/year}$

CHAPTER 7 CONCLUSION AND RECOMMENDATIONS

In this chapter, the conclusion and recommendations are presented based on the researcher's perspective at the prototype development stage. The findings reflect an in-depth analysis of the prototype's performance, highlighting both its strengths and areas for improvement. The conclusions drawn are intended to offer valuable insights into the viability and potential future enhancements of the prototype. Recommendations are provided to guide subsequent development phases, aiming to refine the prototype further, address identified challenges, and optimize its functionality for broader application. This comprehensive overview serves as a foundation for ongoing research and development efforts, ensuring that the project aligns with practical needs and technological advancements.

7.1 Conclusion

The automated PET preform inspection system has truly proven its worth, showcasing impressive performance in key areas like weight measurement, visual defect detection, sensor synchronization, sorting, and conveyor throughput. While we encountered a few minor challenges such as a slight calibration drift in the load cell and some limitations with the ESP32 camera's resolution, we were able to tackle these issues head-on. Through recalibration and thoughtful code optimization, the system demonstrated remarkable adaptability, ensuring reliable operation with both efficiency and accuracy.

This project highlights a promising solution for industrial quality control, particularly within the polymer industry. By enhancing inspection processes, this system can significantly reduce waste and improve product quality, ultimately leading to greater consumer satisfaction. With just a few minor refinements, there is potential for even greater precision and scalability, paving the way for advancements in manufacturing practices and efficiency in the polymer sector. It's exciting to see how technology can enhance processes and contribute to better quality products for consumers.

7.2 Recommendations

Upgrading Camera Resolution:

There is need to replace the ESP32 camera with a higher-resolution model to boost visual defect detection .There is also need to conduct additional experiments at increased conveyor speed to evaluate the system's capacity to maintain accuracy and efficiency under higher production demands.

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CODES

ESP 32 CAMERA CODE

/* Edge Impulse Arduino examples

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*

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* SOFTWARE.

*/

// These sketches are tested with 2.0.4 ESP32 Arduino Core

// <https://github.com/espressif/arduino-esp32/releases/tag/2.0.4>

/* Includes ----- */

#include <TARIE_SORTING_PROJECT_inferencing.h>

#include "edge-impulse-sdk/dsp/image/image.hpp"

#include "esp_camera.h"

// LED indicator pins

#define LED_OK_PIN 13

#define LED_DEFECT_PIN 4

// Select camera model - find more camera models in camera_pins.h file here

// https://github.com/espressif/arduino-esp32/blob/master/libraries/ESP32/examples/Camera/CameraWebServer/camera_pins.h

```
//#define CAMERA_MODEL_ESP_EYE // Has PSRAM

#define CAMERA_MODEL_AI_THINKER // Has PSRAM


#if defined(CAMERA_MODEL_ESP_EYE)

#define PWDN_GPIO_NUM  -1

#define RESET_GPIO_NUM  -1

#define XCLK_GPIO_NUM  4

#define SIOD_GPIO_NUM  18

#define SIOC_GPIO_NUM  23


#define Y9_GPIO_NUM  36

#define Y8_GPIO_NUM  37

#define Y7_GPIO_NUM  38

#define Y6_GPIO_NUM  39

#define Y5_GPIO_NUM  35

#define Y4_GPIO_NUM  14

#define Y3_GPIO_NUM  13

#define Y2_GPIO_NUM  34

#define VSYNC_GPIO_NUM  5

#define HREF_GPIO_NUM  27

#define PCLK_GPIO_NUM  25
```

```
#elif defined(CAMERA_MODEL_AI_THINKER)
```

```
#define PWDN_GPIO_NUM    32
```

```
#define RESET_GPIO_NUM  -1
```

```
#define XCLK_GPIO_NUM    0
```

```
#define SIOD_GPIO_NUM    26
```

```
#define SIOC_GPIO_NUM    27
```

```
#define Y9_GPIO_NUM      35
```

```
#define Y8_GPIO_NUM      34
```

```
#define Y7_GPIO_NUM      39
```

```
#define Y6_GPIO_NUM      36
```

```
#define Y5_GPIO_NUM      21
```

```
#define Y4_GPIO_NUM      19
```

```
#define Y3_GPIO_NUM      18
```

```
#define Y2_GPIO_NUM       5
```

```
#define VSYNC_GPIO_NUM   25
```

```
#define HREF_GPIO_NUM    23
```

```
#define PCLK_GPIO_NUM    22
```

```
#else
```

```
#error "Camera model not selected"
```

```
#endif
```

```
/* Constant defines ----- */
```

```
#define EI_CAMERA_RAW_FRAME_BUFFER_COLS      320
```

```
#define EI_CAMERA_RAW_FRAME_BUFFER_ROWS      240
```

```
#define EI_CAMERA_FRAME_BYTE_SIZE            3
```

```
/* Private variables ----- */
```

```
static bool debug_nn = false;
```

```
static bool is_initialised = false;
```

```
uint8_t *snapshot_buf;
```

```
static camera_config_t camera_config = {
```

```
    .pin_pwdn = PWDN_GPIO_NUM,
```

```
    .pin_reset = RESET_GPIO_NUM,
```

```
    .pin_xclk = XCLK_GPIO_NUM,
```

```
    .pin_sscb_sda = SIOD_GPIO_NUM,
```

```
    .pin_sscb_scl = SIOC_GPIO_NUM,
```

```
    .pin_d7 = Y9_GPIO_NUM,
```

```
    .pin_d6 = Y8_GPIO_NUM,
```

```
    .pin_d5 = Y7_GPIO_NUM,
```

```
    .pin_d4 = Y6_GPIO_NUM,
```

```

.pin_d3 = Y5_GPIO_NUM,

.pin_d2 = Y4_GPIO_NUM,

.pin_d1 = Y3_GPIO_NUM,

.pin_d0 = Y2_GPIO_NUM,

.pin_vsync = VSYNC_GPIO_NUM,

.pin_href = HREF_GPIO_NUM,

.pin_pclk = PCLK_GPIO_NUM,


.xclk_freq_hz = 200000000,

.ledc_timer = LEDC_TIMER_0,

.ledc_channel = LEDC_CHANNEL_0,


.pixel_format = PIXFORMAT_JPEG,

.frame_size = FRAMESIZE_QVGA,


.jpeg_quality = 12,

.fb_count = 1,

.fb_location = CAMERA_FB_IN_PSRAM,

.grab_mode = CAMERA_GRAB_WHEN_EMPTY,
};

/* Function definitions ----- */

```

```

bool ei_camera_init(void);

void ei_camera_deinit(void);

bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t *out_buf);


void setup()

{

    Serial.begin(115200);

    while (!Serial);

    Serial.println("Edge Impulse Inferencing Demo");


    // LED pins setup

    pinMode(LED_OK_PIN, OUTPUT);

    pinMode(LED_DEFECT_PIN, OUTPUT);

    digitalWrite(LED_OK_PIN, LOW);

    digitalWrite(LED_DEFECT_PIN, LOW);


    if (!ei_camera_init()) {

        ei_printf("Failed to initialize Camera!\r\n");

    } else {

        ei_printf("Camera initialized\r\n");

    }
}

```

```

    ei_printf("\nStarting continuous inference in 2 seconds...\n");

    ei_sleep(2000);

}

void loop()

{

    if (ei_sleep(5) != EI_IMPULSE_OK) {

        return;

    }

        snapshot_buf = (uint8_t*)malloc(EI_CAMERA_RAW_FRAME_BUFFER_COLS *
EI_CAMERA_RAW_FRAME_BUFFER_ROWS * EI_CAMERA_FRAME_BYTE_SIZE);

    if (!snapshot_buf) {

        ei_printf("ERR: Failed to allocate snapshot buffer!\n");

        return;

    }

    ei::signal_t signal;

    signal.total_length = EI_CLASSIFIER_INPUT_WIDTH * EI_CLASSIFIER_INPUT_HEIGHT;

    signal.get_data = &ei_camera_get_data;

    if (!ei_camera_capture(EI_CLASSIFIER_INPUT_WIDTH, EI_CLASSIFIER_INPUT_HEIGHT,
snapshot_buf)) {

```

```

    ei_printf("Failed to capture image\r\n");

    free(snapshot_buf);

    return;
}

ei_impulse_result_t result = { 0 };

EI_IMPULSE_ERROR err = run_classifier(&signal, &result, debug_nn);

if (err != EI_IMPULSE_OK) {

    ei_printf("ERR: Failed to run classifier (%d)\n", err);

    free(snapshot_buf);

    return;
}

ei_printf("Predictions (DSP: %d ms., Classification: %d ms., Anomaly: %d ms.): \n",

        result.timing.dsp, result.timing.classification, result.timing.anomaly);

bool saw_defective = false;

bool saw_not_defective = false;

#if EI_CLASSIFIER_OBJECT_DETECTION == 1

    ei_printf("Object detection bounding boxes:\r\n");

    for (uint32_t i = 0; i < result.bounding_boxes_count; i++) {

```

```

    auto &bb = result.bounding_boxes[i];

    if (bb.value <= 0) continue;

    ei_printf(" %s (%f) [ x: %u, y: %u, width: %u, height: %u ]\r\n",

        bb.label, bb.value, bb.x, bb.y, bb.width, bb.height);

    if (strcmp(bb.label, "DEFECTIVE") == 0 && bb.value > 0.5f) saw_defective = true;

    if (strcmp(bb.label, "NOT_DEFECTIVE") == 0 && bb.value > 0.5f) saw_not_defective = true;

}

#else

ei_printf("Predictions:\r\n");

for (uint16_t i = 0; i < EI_CLASSIFIER_LABEL_COUNT; i++) {

    auto &c = result.classification[i];

    ei_printf(" %s: %.5f\r\n", ei_classifier_inferencing_categories[i], c.value);

    if (strcmp(ei_classifier_inferencing_categories[i], "DEFECTIVE") == 0 && c.value > 0.5f)
saw_defective = true;

    if (strcmp(ei_classifier_inferencing_categories[i], "NOT_DEFECTIVE") == 0 && c.value > 0.5f)
saw_not_defective = true;

}

#endif

// Set LEDs: defective has priority

if (saw_defective) {

    digitalWrite(LED_DEFECT_PIN, HIGH);

    digitalWrite(LED_OK_PIN, LOW);

```

```

    } else if (saw_not_defective) {

        digitalWrite(LED_OK_PIN, HIGH);

        digitalWrite(LED_DEFECT_PIN, LOW);

    } else {

        // no confident result

        digitalWrite(LED_OK_PIN, LOW);

        digitalWrite(LED_DEFECT_PIN, LOW);

    }

    free(snapshot_buf);

}

bool ei_camera_init(void) {

    if (is_initialised) return true;

#ifdef CAMERA_MODEL_ESP_EYE

    pinMode(13, INPUT_PULLUP);

    pinMode(14, INPUT_PULLUP);

#endif

    esp_err_t err = esp_camera_init(&camera_config);

    if (err != ESP_OK) {

        Serial.printf("Camera init failed with error 0x%x\n", err);

        return false;
    }

```

```

    }

    sensor_t *s = esp_camera_sensor_get();

    if (s->id.PID == OV3660_PID) {

        s->set_vflip(s, 1);

        s->set_brightness(s, 1);

        s->set_saturation(s, 0);

    }

#ifdef CAMERA_MODEL_M5STACK_WIDE

    s->set_vflip(s, 1);

    s->set_hmirror(s, 1);

#elif defined(CAMERA_MODEL_ESP_EYE)

    s->set_vflip(s, 1);

    s->set_hmirror(s, 1);

    s->set_awb_gain(s, 1);

#endif

    is_initialised = true;

    return true;

}

void ei_camera_deinit(void) {

    esp_err_t err = esp_camera_deinit();

    if (err != ESP_OK) {

```

```

        ei_printf("Camera deinit failed\n");

    }

    is_initialised = false;

}

bool ei_camera_capture(uint32_t img_width, uint32_t img_height, uint8_t *out_buf) {

    if (!is_initialised) {

        ei_printf("ERR: Camera is not initialized\r\n");

        return false;

    }

    camera_fb_t *fb = esp_camera_fb_get();

    if (!fb) {

        ei_printf("Camera capture failed\n");

        return false;

    }

    bool converted = fmt2rgb888(fb->buf, fb->len, PIXFORMAT_JPEG, snapshot_buf);

    esp_camera_fb_return(fb);

    if (!converted) {

        ei_printf("Conversion failed\n");

        return false;

    }

```

```

        if (img_width != EI_CAMERA_RAW_FRAME_BUFFER_COLS || img_height !=
EI_CAMERA_RAW_FRAME_BUFFER_ROWS) {

    ei::image::processing::crop_and_interpolate_rgb888(

        out_buf,

        EI_CAMERA_RAW_FRAME_BUFFER_COLS,

        EI_CAMERA_RAW_FRAME_BUFFER_ROWS,

        out_buf,

        img_width,

        img_height);

    }

    return true;

}

```

```

static int ei_camera_get_data(size_t offset, size_t length, float *out_ptr) {

    size_t pixel_ix = offset * 3;

    size_t pixels_left = length;

    size_t out_ptr_ix = 0;

    while (pixels_left--) {

        out_ptr[out_ptr_ix++] = (snapshot_buf[pixel_ix+2]<<16) + (snapshot_buf[pixel_ix+1]<<8) +
snapshot_buf[pixel_ix];

        pixel_ix += 3;

    }

    return 0;
}

```

```
}
```

```
#if !defined(EI_CLASSIFIER_SENSOR) || EI_CLASSIFIER_SENSOR !=  
EI_CLASSIFIER_SENSOR_CAMERA  
  
#error "Invalid model for current sensor"  
  
#endif
```

ESP 32 MICRO CONTROLLER

```
#include <HX711.h>  
  
#include <ESP32Servo.h>  
  
#include <Arduino.h>  
  
#include <Wire.h>  
  
#include <LiquidCrystal_I2C.h>
```

```

// ----- PINS -----

#define ENA      32  // Motor A PWM

#define ENB      33  // Motor B PWM

#define IN1      25  // Motor A dir

#define IN2      26

#define IN3      27  // Motor B dir

#define IN4      14  // Motor B dir


// HX711 Scale

#define LOADCELL_DOUT_PIN 16

#define LOADCELL_SCK_PIN 17

HX711 scale;

const float calibration_factor = -1168000.0; // your recalibrated factor


// Servos

#define SERVO_DEFECT_PIN 4

#define SERVO_OK_PIN    5


// Buttons

#define BUTTON_DEFECT_PIN 18

#define BUTTON_OK_PIN    19

```

```

// LEDs

#define LED_BLUE_PIN 23

#define LED_GREEN_PIN 13

#define LED_RED_PIN 12


// ESP32-CAM auto-decision inputs

#define PIN_DEFECT_IN 15

#define PIN_OK_IN 35


// Ultrasonic sensor

const int trigPin = 2;

const int echoPin = 34;

const int NUM_SAMPLES = 7;

const float SOUND_FACTOR = 0.0343 / 2.0;

typedef unsigned long ul;

const float OBJ_TH = 10.0;

const float CLEAR_TH = 15.0;

const int CONFIRM_COUNT = 3;

long samples[NUM_SAMPLES];

int consecCount = 0;

bool ultrasonObject = false;

const ul ULTRA_INTERVAL = 100;

```

```

ul lastUltraMillis    = 0;

// Motor & timing constants

const int pwmFreq      = 5000;

const int pwmResolution = 8;

enum { REF_SPEED = 100,

      RUN_SPEED = (REF_SPEED * 85) / 100,

      BOOST_SPEED = REF_SPEED };

const ul MOTOR_BOOST_MS = 200;

const ul SERVO_HOLD_MS = 3000;

const ul DEBOUNCE_MS   = 50;

const ul BLUE_BLINK_MS = 500;

// Auto-decision sampling

const unsigned long WINDOW_MS = 100;

const unsigned long SAMPLE_INTERVAL_MS = 10;

const int REQUIRED_CONSECUTIVE_WINDOWS = 3;

unsigned long windowStart = 0;

int okCount = 0, defectCount = 0;

enum Decision { NONE, OK_DECISION, DEFECT_DECISION };

Decision lastWindowDecision = NONE, finalDecision = NONE;

int consecutiveWindowCount = 0;

```

```

// PC LCD (16×2)

#define LCD_ADDR 0x27

LiquidCrystal_I2C lcd(LCD_ADDR, 16, 2);


// Globals

Servo servoDefect, servoOK;

enum State { RUNNING, MEASUREMENT, WAIT_BUTTON, SERVO_WAIT };

State state = RUNNING;

bool isDefectSelected = false;

ul actionStartTime = 0;

bool motorBoostActive = false;

ul motorBoostStartTime = 0;


// Button debounce

int lastRawDef = LOW, debouncedDef = LOW;

ul lastDebounceDef = 0; bool justPressedDef = false;

int lastRawOk = LOW, debouncedOk = LOW;

ul lastDebounceOk = 0; bool justPressedOk = false;


// LED blink tracking

ul lastBlueToggle = 0;

```

```
bool blueOn      = false;
```

```
//————— Helpers —————
```

```
void updateButton(int pin, int &lastRaw, int &debounced,
```

```
    ul &lastDebounceTime, bool &justPressed) {
```

```
    int raw = digitalRead(pin);
```

```
    if (raw != lastRaw) lastDebounceTime = millis();
```

```
    if (millis() - lastDebounceTime >= DEBOUNCE_MS) {
```

```
        if (raw != debounced) {
```

```
            debounced = raw;
```

```
            if (debounced == HIGH) justPressed = true;
```

```
        }
```

```
    }
```

```
    lastRaw = raw;
```

```
}
```

```
float getMedianDistance() {
```

```
    for (int i = 0; i < NUM_SAMPLES; i++) {
```

```
        digitalWrite(trigPin, LOW);
```

```
        delayMicroseconds(2);
```

```
        digitalWrite(trigPin, HIGH);
```

```
        delayMicroseconds(10);
```

```

digitalWrite(trigPin, LOW);

samples[i] = pulseIn(echoPin, HIGH, 30000UL);

delay(5);

}

// Insertion sort for median

for (int i = 1; i < NUM_SAMPLES; i++) {

    long key = samples[i];

    int j = i - 1;

    while (j >= 0 && samples[j] > key) {

        samples[j + 1] = samples[j];

        j--;

    }

    samples[j + 1] = key;

}

return samples[NUM_SAMPLES/2] * SOUND_FACTOR;

}

void updateUltrasonic() {

    ul now = millis();

    if (now - lastUltraMillis < ULTRA_INTERVAL) return;

    lastUltraMillis = now;

    float dist = getMedianDistance();

```

```

if (!ultrasonObject) {

    if (dist > 0 && dist < OBJ_TH && ++consecCount >= CONFIRM_COUNT) {

        ultrasonObject = true; consecCount = 0;

    } else if (dist >= OBJ_TH) consecCount = 0;

    } else {

        if (dist > CLEAR_TH && ++consecCount >= CONFIRM_COUNT) {

            ultrasonObject = false; consecCount = 0;

        } else if (dist <= CLEAR_TH) consecCount = 0;

    }

}

```

```

void setup() {

    Serial.begin(115200);

    // Motor pins

    pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);

    pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);

    pinMode(ENA, OUTPUT); pinMode(ENB, OUTPUT);

    analogWriteFrequency(ENA, pwmFreq);

    analogWriteResolution(ENA, pwmResolution);

    analogWriteFrequency(ENB, pwmFreq);

    analogWriteResolution(ENB, pwmResolution);

```

```

digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);

digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);


// Scale init & tare

scale.begin(LOADCELL_DOUT_PIN, LOADCELL_SCK_PIN);

scale.set_scale(calibration_factor);

scale.tare();

Serial.println("Scale ready");


// Servos

servoDefect.attach(SERVO_DEFECT_PIN);

servoOK.attach(SERVO_OK_PIN);

servoDefect.write(0);

servoOK.write(120);


// Buttons

pinMode(BUTTON_DEFECT_PIN, INPUT_PULLDOWN);

pinMode(BUTTON_OK_PIN, INPUT_PULLDOWN);


// Auto-decision inputs

pinMode(PIN_DEFECT_IN, INPUT_PULLUP);

pinMode(PIN_OK_IN, INPUT);

```

```
windowStart = millis();

// LEDs

pinMode(LED_BLUE_PIN, OUTPUT);

pinMode(LED_GREEN_PIN, OUTPUT);

pinMode(LED_RED_PIN, OUTPUT);

lastBlueToggle = millis();

blueOn = false;


// Ultrasonic

pinMode(trigPin, OUTPUT);

pinMode(echoPin, INPUT);


// LCD splash

Wire.begin(21, 22);

lcd.begin();

lcd.backlight();

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("TARIRO");

lcd.setCursor(0, 1);

lcd.print("SORTING_PROJECT");
```

```
}
```

```
void loop() {
```

```
    // Debounce buttons
```

```
    justPressedDef = justPressedOk = false;
```

```
        updateButton(BUTTON_DEFECT_PIN,    lastRawDef,    debouncedDef,    lastDebounceDef,
justPressedDef);
```

```
    updateButton(BUTTON_OK_PIN,    lastRawOk,    debouncedOk,    lastDebounceOk,    justPressedOk);
```

```
    // Auto-decision sampling
```

```
    ul now = millis();
```

```
    static ul lastSample = 0;
```

```
    if (now - lastSample >= SAMPLE_INTERVAL_MS) {
```

```
        lastSample = now;
```

```
        if (digitalRead(PIN_OK_IN))    okCount++;
```

```
        if (digitalRead(PIN_DEFECT_IN)) defectCount++;
```

```
    }
```

```
    if (now - windowStart >= WINDOW_MS) {
```

```
        Decision windowDec = NONE;
```

```
        if (defectCount > okCount && defectCount - okCount >= 2) windowDec = DEFECT_DECISION;
```

```
        else if (okCount > defectCount && okCount - defectCount >= 2) windowDec = OK_DECISION;
```

```

if (windowDec == lastWindowDecision && windowDec != NONE)

    consecutiveWindowCount++;

else {

    lastWindowDecision = windowDec;

    consecutiveWindowCount = (windowDec==NONE ? 0 : 1);

}

        if    (windowDec    !=    NONE    &&    consecutiveWindowCount    >=
REQUIRED_CONSECUTIVE_WINDOWS)

    finalDecision = windowDec;


    okCount = defectCount = 0;

    windowStart = now;

}


// Ultrasonic sensor

updateUltrasonic();


// State machine

switch (state) {

    case RUNNING:

        if (ultrasonObject) {

            analogWrite(ENA, 0);

```

```

    analogWrite(ENB, 0);

    Serial.println("STOPPED: Object detected!");

    state = MEASUREMENT;

} else {

    int speed = (motorBoostActive && now - motorBoostStartTime < MOTOR_BOOST_MS)

        ? BOOST_SPEED : RUN_SPEED;

    analogWrite(ENA, speed);

    analogWrite(ENB, speed);

    Serial.print("Moving forward at ");

    Serial.print(speed);

    Serial.println("%");

}

break;

case MEASUREMENT: {

    // Take 10 readings & average

    const int N = 10;

    float sum = 0;

    delay(200);

    for (int i = 0; i < N; i++) {

        sum += scale.get_units(1);

        delay(50);

```

```
}
```

```
float weight = sum / N; // in kg
```

```
Serial.print("Weight: ");
```

```
Serial.print(weight, 4);
```

```
Serial.println(" kg");
```

```
// Display on LCD
```

```
lcd.clear();
```

```
lcd.setCursor(0, 0);
```

```
lcd.print("WEIGHT:");
```

```
lcd.setCursor(8, 0);
```

```
lcd.print(weight, 4);
```

```
lcd.print("kg");
```

```
state = WAIT_BUTTON;
```

```
break;
```

```
}
```

```
case WAIT_BUTTON:
```

```
if (justPressedDef)    isDefectSelected = true;
```

```
else if (justPressedOk) isDefectSelected = false;
```

```

else if (finalDecision != NONE)

    isDefectSelected = (finalDecision == DEFECT_DECISION);

else

    break;


// Reset auto-decision

finalDecision = NONE;

lastWindowDecision = NONE;

consecutiveWindowCount = 0;


// Boost belt

motorBoostActive = true;

motorBoostStartTime = now;

analogWrite(ENA, BOOST_SPEED);

analogWrite(ENB, BOOST_SPEED);


// Servo + LCD message

if (isDefectSelected) {

    Serial.println("Selected: DEFECTIVE");

    servoDefect.write(120);

    lcd.setCursor(0, 1);

    lcd.print("DEFECTIVE  ");

```

```

} else {

    Serial.println("Selected: NOT DEFECTIVE");

    servoOK.write(0);

    lcd.setCursor(0, 1);

    lcd.print("NOT DEFECTIVE ");

}

```

```

actionStartTime = now;

state = SERVO_WAIT;

break;

```

```

case SERVO_WAIT:

    if (now - actionStartTime >= SERVO_HOLD_MS) {

        if (isDefectSelected) servoDefect.write(0);

        else                servoOK.write(120);

        // Restore splash

        lcd.clear();

        lcd.setCursor(0, 0);

        lcd.print("TARIRO");

        lcd.setCursor(0, 1);

        lcd.print("SORTING_PROJECT");

```

```

        state = RUNNING;

    }

    break;

}

// LEDs

bool motorRunning = (state == RUNNING) || (state == SERVO_WAIT);

if (motorRunning) {

    digitalWrite(LED_BLUE_PIN, HIGH);

    blueOn = true;

    lastBlueToggle = now;

} else if (now - lastBlueToggle >= BLUE_BLINK_MS) {

    blueOn = !blueOn;

    lastBlueToggle = now;

}

digitalWrite(LED_BLUE_PIN, blueOn);

digitalWrite(LED_GREEN_PIN, (state == RUNNING));

digitalWrite(LED_RED_PIN, (state != RUNNING));

}

```

